

Volatility Skews

There are clearly real problems associated with the use of a traditional theoretical pricing model. Markets are not frictionless, prices do not always follow a diffusion process, volatility may vary over the life of an option, the real world may not look like a lognormal distribution. With all these weaknesses, one might wonder whether theoretical pricing models have any practical value at all. In fact, most traders have found that pricing models, while not perfect, are an invaluable tool for making decisions in the option market. Even if a model does not work perfectly, traders have found that using a model, even a flawed one, is usually better than using no model at all.

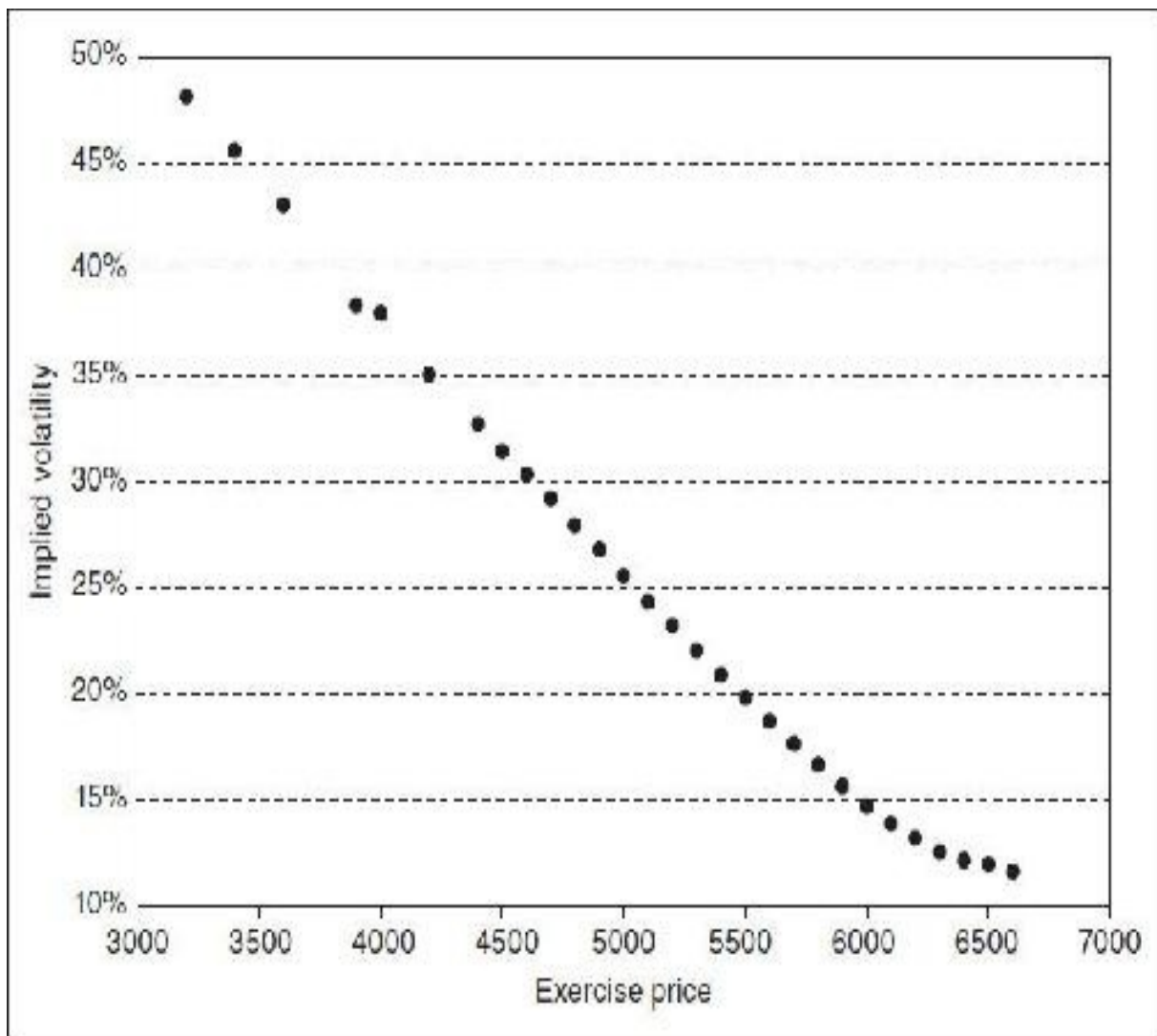
Still, a trader who wants to make the best possible decisions cannot afford to ignore the problems associated with a theoretical pricing model. Consequently, a trader who uses a pricing model might look for a way to reduce the potential errors resulting from these weaknesses. Initially, one might simply look for a better theoretical pricing model. If such a model exists, it will certainly be worth replacing the old model with the new one. But *better* is a relative term. A model might be better in the sense that it gives slightly more accurate theoretical values. But if the model is extremely complex and difficult to use, or if it requires additional inputs of which a trader cannot always be certain, then the model may merely substitute one set of problems for another. Given the fact that most traders are not theoreticians, a more realistic solution might be to use a less complex model and somehow fine-tune it so that it is consistent with the realities of the marketplace.

A trader trying to compensate for weaknesses in a pricing model might make the assumption that the marketplace is using the same model as the trader and then ask how the marketplace is dealing with the weaknesses in the model. This is somewhat analogous to calculating implied volatility where we assume that everyone is using the same model, that the price of the option is known, and that everyone agrees on all the inputs except volatility. From these assumptions, we are able to determine the volatility that the marketplace is implying to the underlying contract. We can take the same general approach but ask instead what

weaknesses the marketplace is implying to the model.

[Figure 24-1](#) shows the implied volatilities across exercise prices for June 2012 FTSE 100 Index¹ options traded on the London International Financial Exchange on March 16, 2012. Calculations were made at the end of the trading day from the average of the bid-ask spread using the Black-Scholes model. It is immediately apparent that implied volatilities vary across exercise prices. If we assume that the exercise price, time to expiration, underlying price, and interest rate are known, the theoretical value of an option in a Black-Scholes world will depend solely on the volatility of the underlying contract over the life of the option. Of course, we won't know what that volatility is until we reach expiration, at which time we can look back and calculate the historical volatility over the 13-week period from March 16 to June expiration. But the FTSE 100 Index can have only one volatility over this period. Because the underlying index is the same for all options, it doesn't make sense in a perfect Black-Scholes world for every exercise price to have a different implied volatility. If the activity in the marketplace were a result of everyone believing in the efficiency of the Black-Scholes model, the selling of overpriced options and the buying of underpriced options would eventually cause every option to have the same implied volatility. Yet this almost never happens in any market.

Figure 24-1 June 2012 FTSE 100 implied volatilities: March 16, 2012.



The distribution of implied volatilities across exercise prices is often referred to as a *volatility skew* or, possibly, a *volatility smile* or *volatility smirk* depending on the shape of the skew. One likely explanation for the distribution of implied volatilities has to do with the way in which options are used as a hedging instrument. In the stock market, most investors are long stock² and are therefore concerned about a decline in stock prices. The two most common hedging strategies to protect a long underlying position, as described in [Chapter 17](#), are the purchase of protective puts and the sale of covered calls.

If a stock investor decides to purchase a protective put, which exercise price will he choose? An out-of-the-money put costs less than an in-the-money put but also offers less protection against a down move. However, if the investor is so worried about a downward move that he needs the protection afforded by an in-the-money put, he ought to simply sell the stock. The result is that most

protective puts are purchased at lower exercise prices.

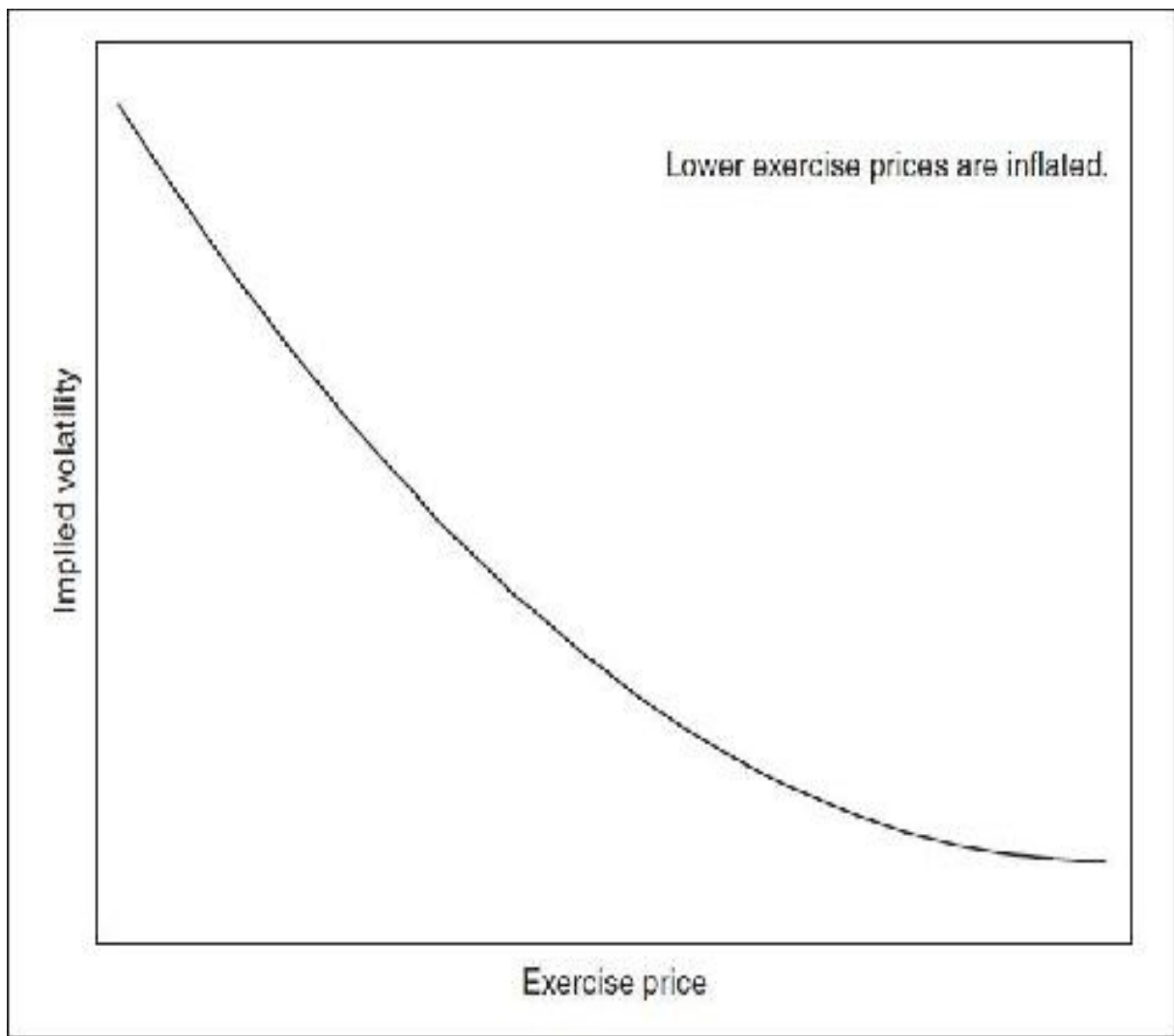
If, instead, the investor decides to sell a covered call, he will almost always do so at a higher exercise price. This will offer less protection than the sale of an in-the-money call, but presumably the investor holds the stock because he believes that the stock price will rise. If it does rise, he will want to participate in at least some of the upside profit potential. If the stock rises and the investor has sold an in-the-money call, the stock will be quickly called away, limiting any upside profit. The result is that most covered calls are sold at higher exercise prices.

As a result of hedging activity, in the stock option market there tends to be buying pressure on the lower exercise prices (the purchase of protective puts) and selling pressure on the higher exercise prices (the sale of covered calls). This causes the implied volatilities at lower exercise prices to rise and the implied volatilities at higher exercise prices to fall. The resulting skew, such as that in [Figure 24-1](#), is sometimes referred to as an *investment skew*. It occurs in markets in which people freely invest, the most obvious example being the stock market. Traders sometimes describe an investment skew by saying that the “skew is to the puts,” indicating that put implied volatilities are inflated. But put-call parity dictates that if a put price is inflated, the call price at the same exercise price must also be inflated, so perhaps it is more accurate to say that the “skew is to the downside.”

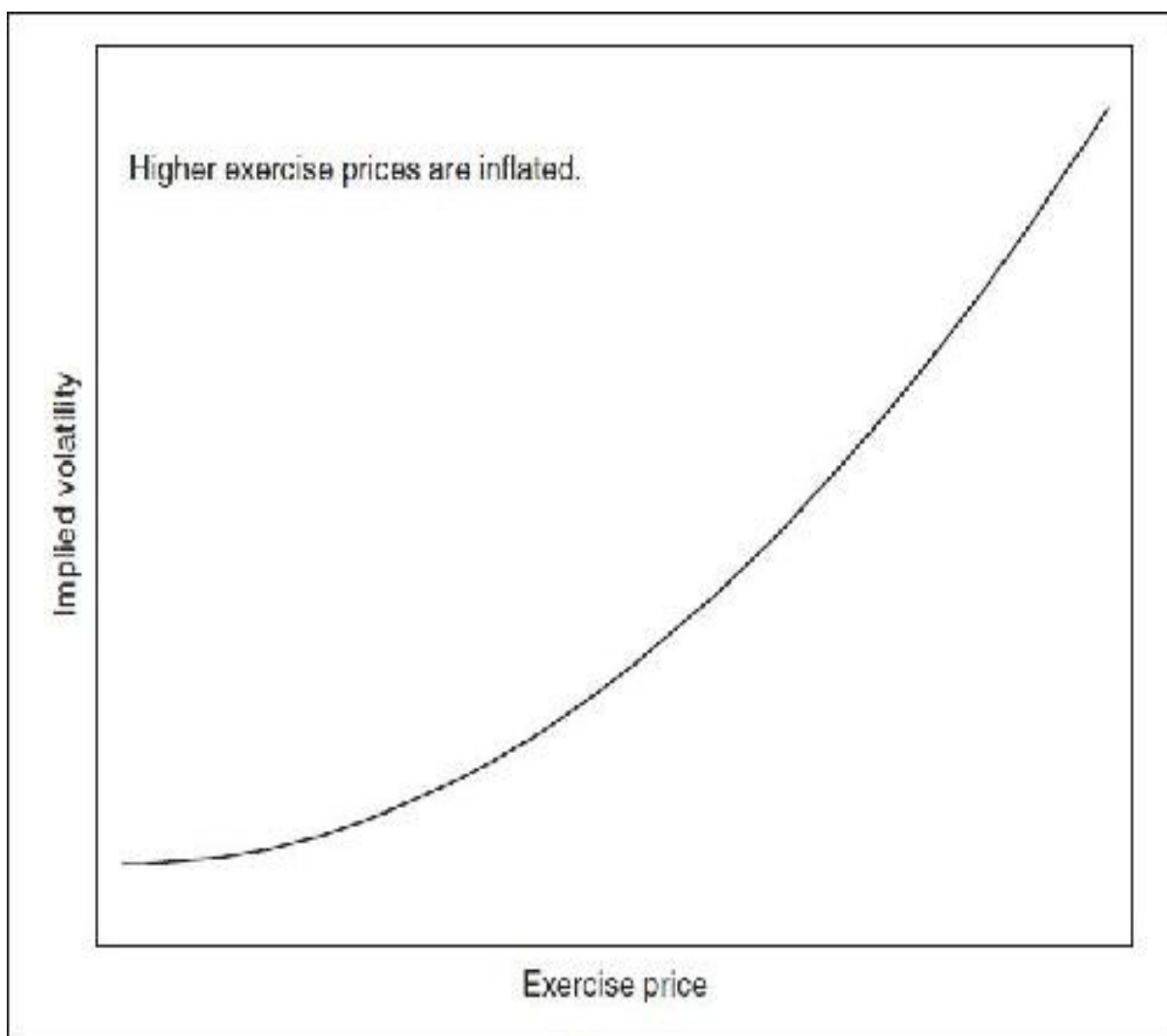
While investors in the stock market may worry about falling stock prices, in other markets hedgers may worry about rising prices. This is often the case in commodity markets where end users try to protect themselves against rising prices by either buying protective calls at higher exercise prices or selling covered puts at lower exercise prices. In the resulting *demand* or *commodity skew* (there is a demand for the commodity), lower exercise prices have lower implied volatilities, and higher exercise prices have higher implied volatilities. Of course, commodity producers, such as farmers, mining companies, and oil drilling companies, are likely to worry about falling commodity prices, so it might seem that there ought to be equal hedging activity between the longs (producers) and the shorts (end users). But in many markets the end users tend to dominate, perhaps because higher commodity prices, and the concomitant inflationary pressures, are perceived as having a negative effect on the entire economy. Moreover, in some countries, the government has a program of price supports for agricultural products, so growers have less to worry about from falling agricultural commodity prices than end users have from rising prices.

Finally, there are markets where both longs and shorts are equally worried. Consider a U.S. company buying goods in Europe that must be paid for on some future date in euros. The company clearly is worried about a rising euro compared with the dollar. At the same time, a European company may buy goods in the United States that must be paid for in dollars. This company is worried about a falling euro compared with the dollar. If both companies choose to hedge their risk in the currency option market, the hedging activity will tend to result in a *balanced skew*, where there is no obvious domination of implied volatilities at either higher or lower exercise prices. This does not mean that implied volatilities will necessarily form a *flat skew*, but the distribution of implied volatilities is likely to be symmetrical around the current underlying price. The three common types of skews are shown in [Figure 24-2](#).

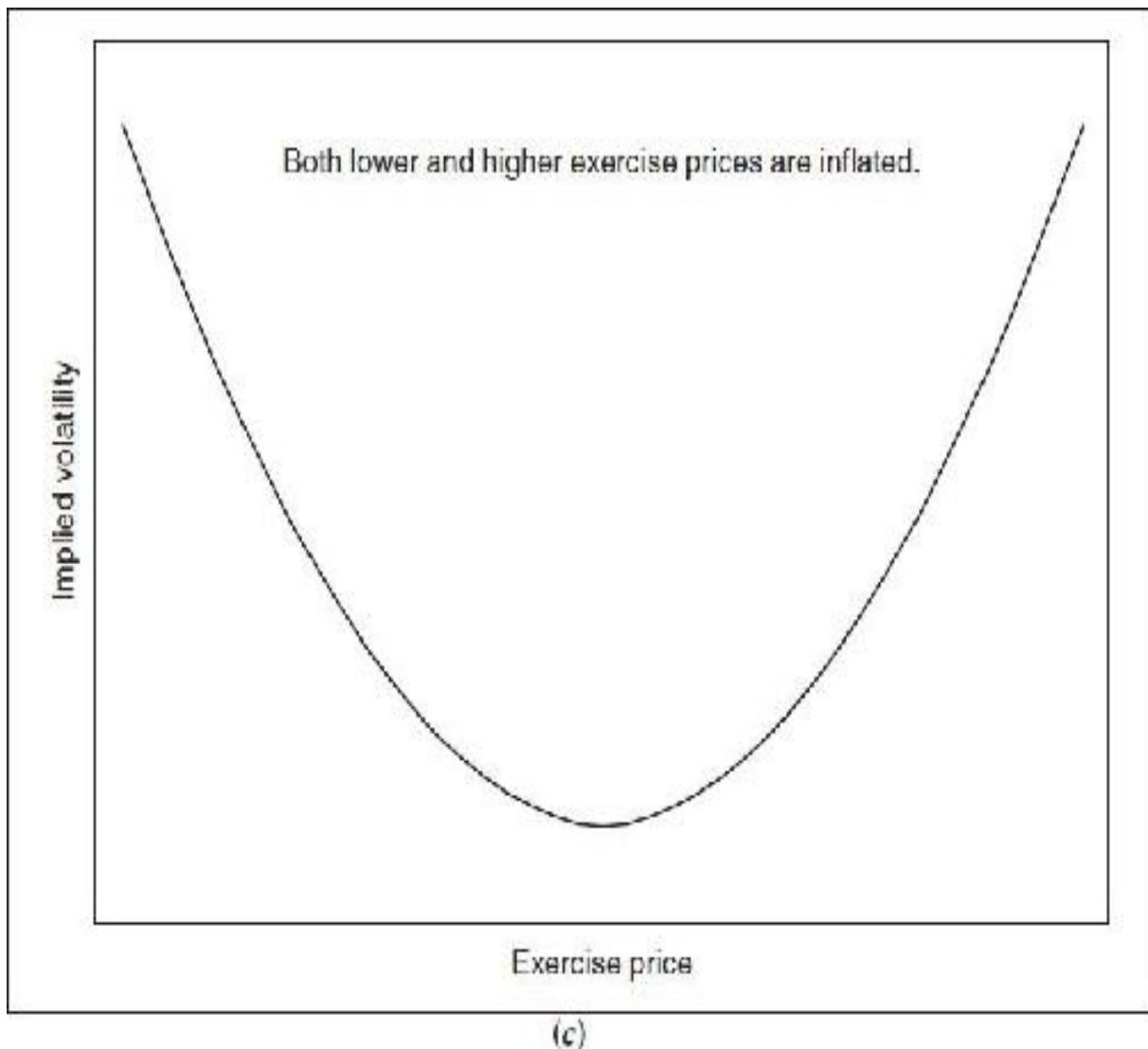
Figure 24-2 (a) Investment skew. (b) Demand skew. (c) Balanced skew.



(a)



(b)



In addition to distortions caused by hedging activity, we also know from [Chapter 23](#) that there are inherent weaknesses in many models. For example, most traders believe that stock markets become more volatile when they are falling and less volatile when they are rising. We also know that an option is most sensitive to volatility changes (it has its highest vega) when it is at the money. If an underlying stock is trading at 100 and the market begins to fall, the vega of the 95 put will rise because it is becoming more at the money. If the market also becomes more volatile because the stock price is falling, this will increase the volatility value of the 95 put. But, if the market begins to rise, the 105 call, even though its vega is rising, will not benefit to the same extent as the 95 put because the market is becoming less volatile. So it should not come as a surprise that the 95 put carries a higher implied volatility and the 105 call a

lower implied volatility than expected. This is consistent with an investment skew.

The marketplace, like every individual trader, is trying to evaluate options as efficiently as possible given all available information. Whether one believes that markets are efficient or not, one can argue that the marketplace is trying to be efficient. From the wide range of implied volatilities found in almost every option market, we can reasonably infer that the marketplace does not think the Black-Scholes model is perfectly efficient. Unfortunately, trying to identify the source of the inefficiency may not be possible. It might have to do with how options are used in hedging strategies. Or it might have to do with weaknesses in the theoretical pricing model. Whatever the reason, we can make the assumption that at any moment in time the marketplace believes that options are priced efficiently, even if those prices happen to differ from model-generated values.

An option trader using a theoretical pricing model might take the view that the volatility skew contains useful information that can be used in the decision-making process. By treating the volatility skew as an additional input into the theoretical pricing model, the skew becomes an important aid in generating theoretical values and managing risk. Moreover, analysis of the skew can form the basis for a variety of option strategies.

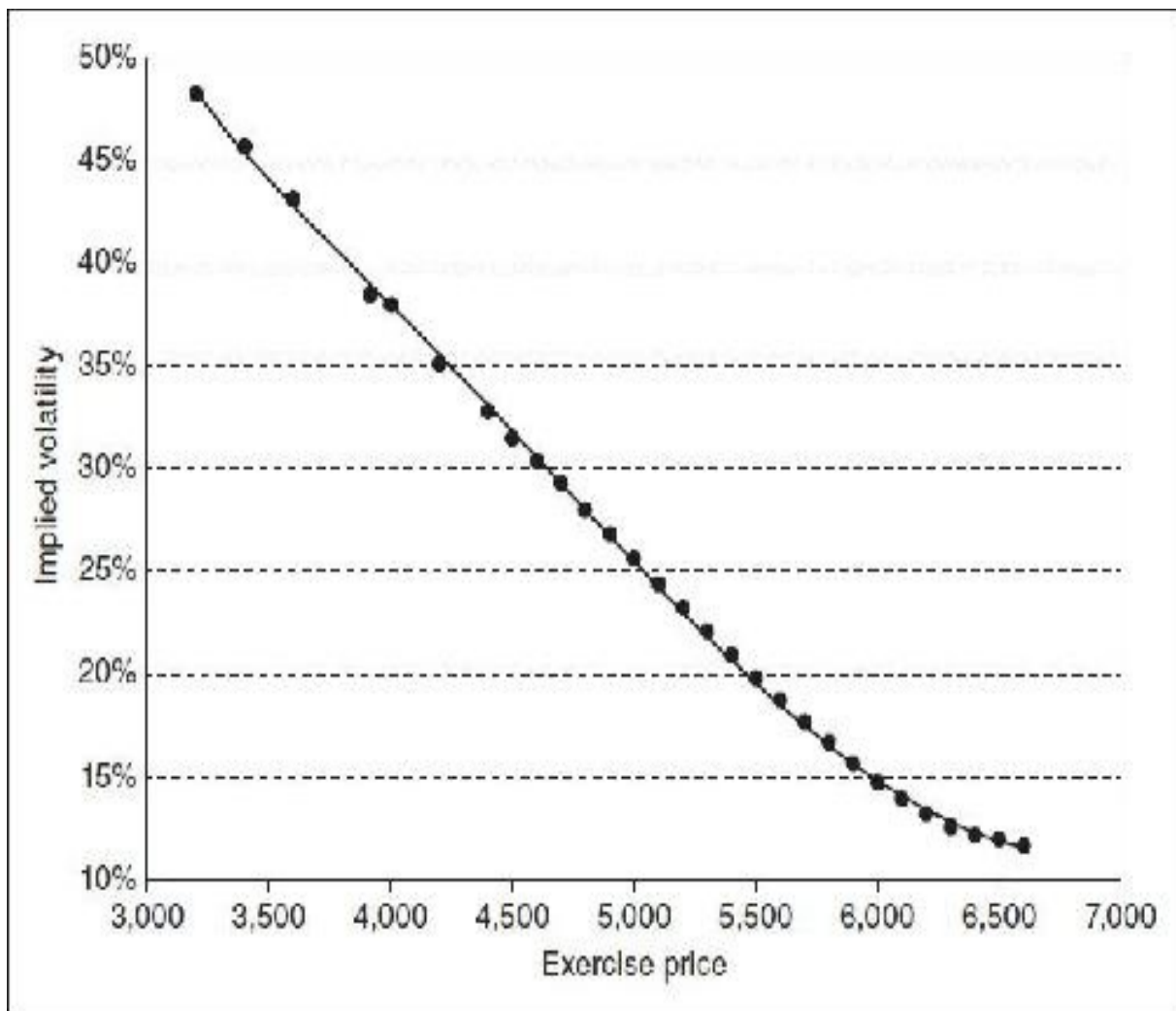
Modeling the Skew

If we want to include a skew in our model, we need to do it in a way that the model understands. This is typically done using a mathematical function that generates a best fit for the skew

$$f(x) = y$$

where y is the implied volatility at each exercise price x . A trader can choose any function that seems to yield a good fit, but many traders use a polynomial function of the form $a + bx + cx^2 + dx^3 + \dots$. A best-fit function for the implied volatilities in [Figure 24-1](#) is shown in [Figure 24-3](#).

Figure 24-3 June 2012 FTSE 100 implied volatilities: March 16, 2012.



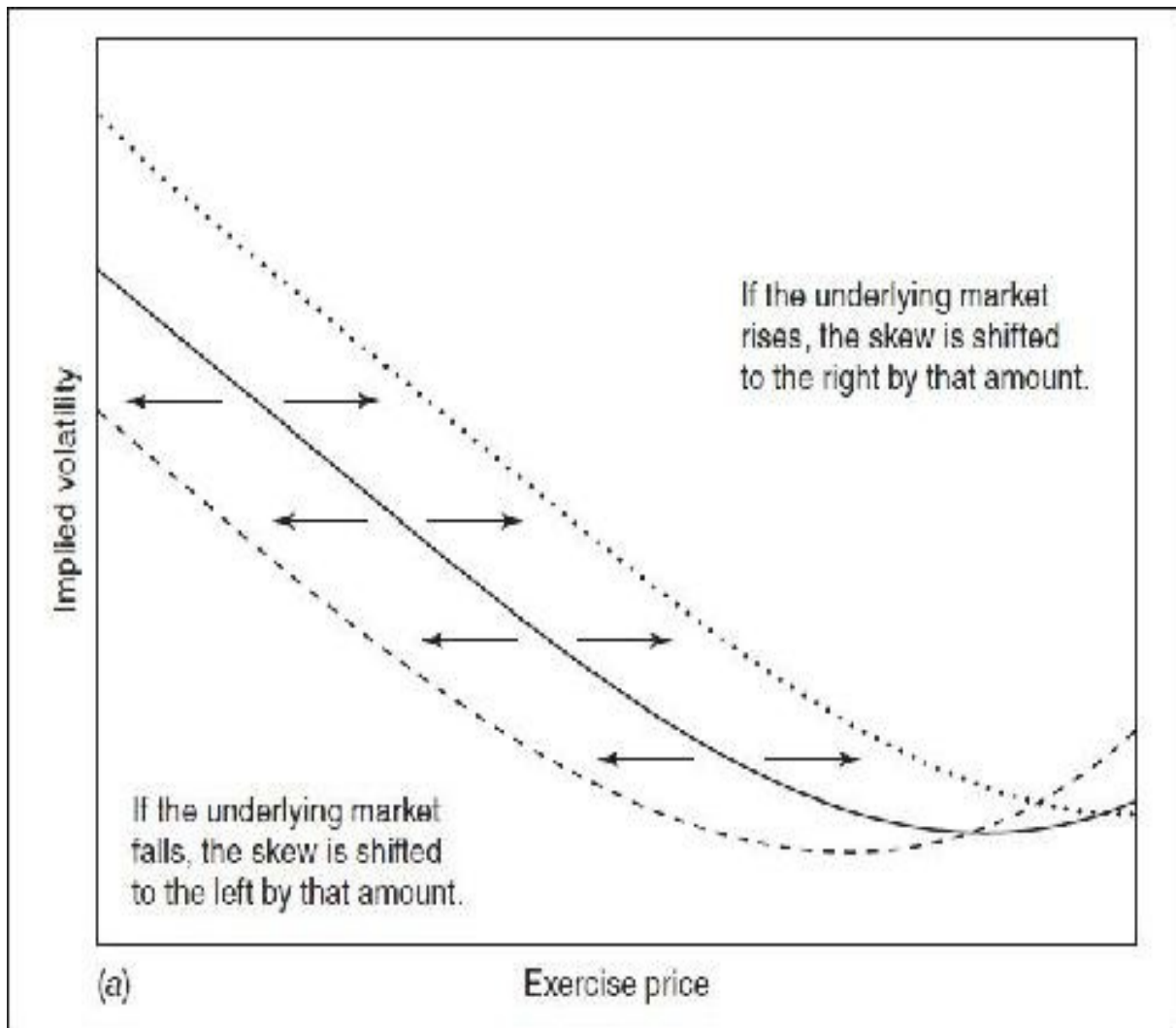
If we think of the skew as an input into the model, then, as with all inputs, we need to ask how changes in the input will affect a position. If we can model possible changes in the skew as market conditions change, we will be in a better position to assess the risk associated with an option position. In particular, as market conditions change, we will want to model both the location and shape of the skew.

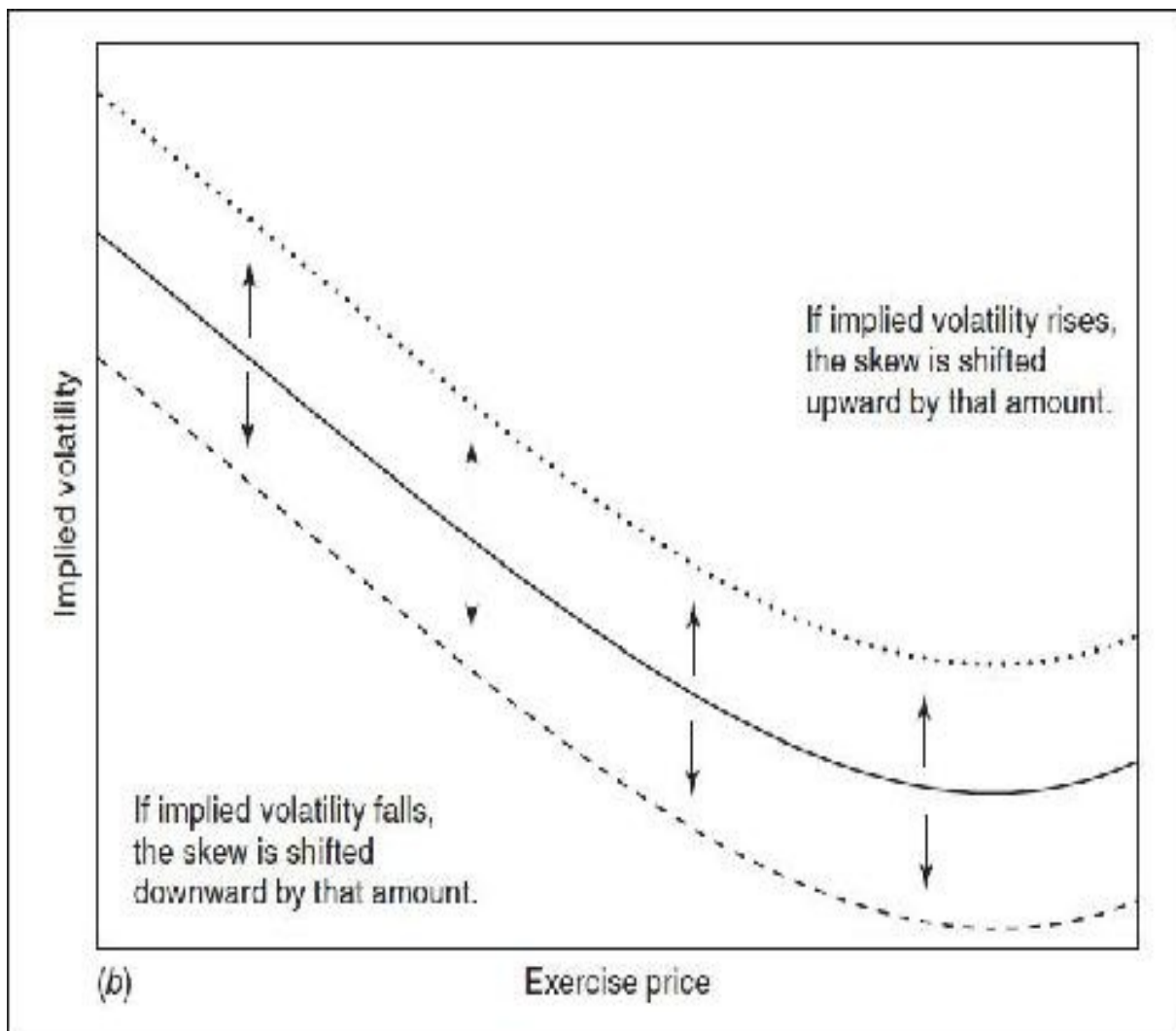
Of course, we might take the position that the location and shape of the volatility skew will remain fixed. Under this *sticky-strike* assumption, the current skew determines the implied volatility at each strike regardless of how market conditions change.

Unfortunately, a sticky-strike skew, with its fixed volatilities at each exercise price, is not consistent with the observed dynamics of the marketplace. In most option markets, the skew will shift as the underlying price moves or implied volatility changes. An alternative approach is to use a *floating skew*,

where the entire skew is shifted horizontally as the underlying price rises or falls or vertically as implied volatility rises or falls. The shift is equal to the amount of change in either the price or volatility. If the underlying price rises five points, the skew is shifted to the right by five points. If implied volatility falls by two percentage points, the skew is shifted downward by two percentage points. This type of skew is shown in [Figure 24-4](#).

Figure 24-4 (a) A simple floating skew as the underlying price changes. (b) A simple floating skew as implied volatility changes.





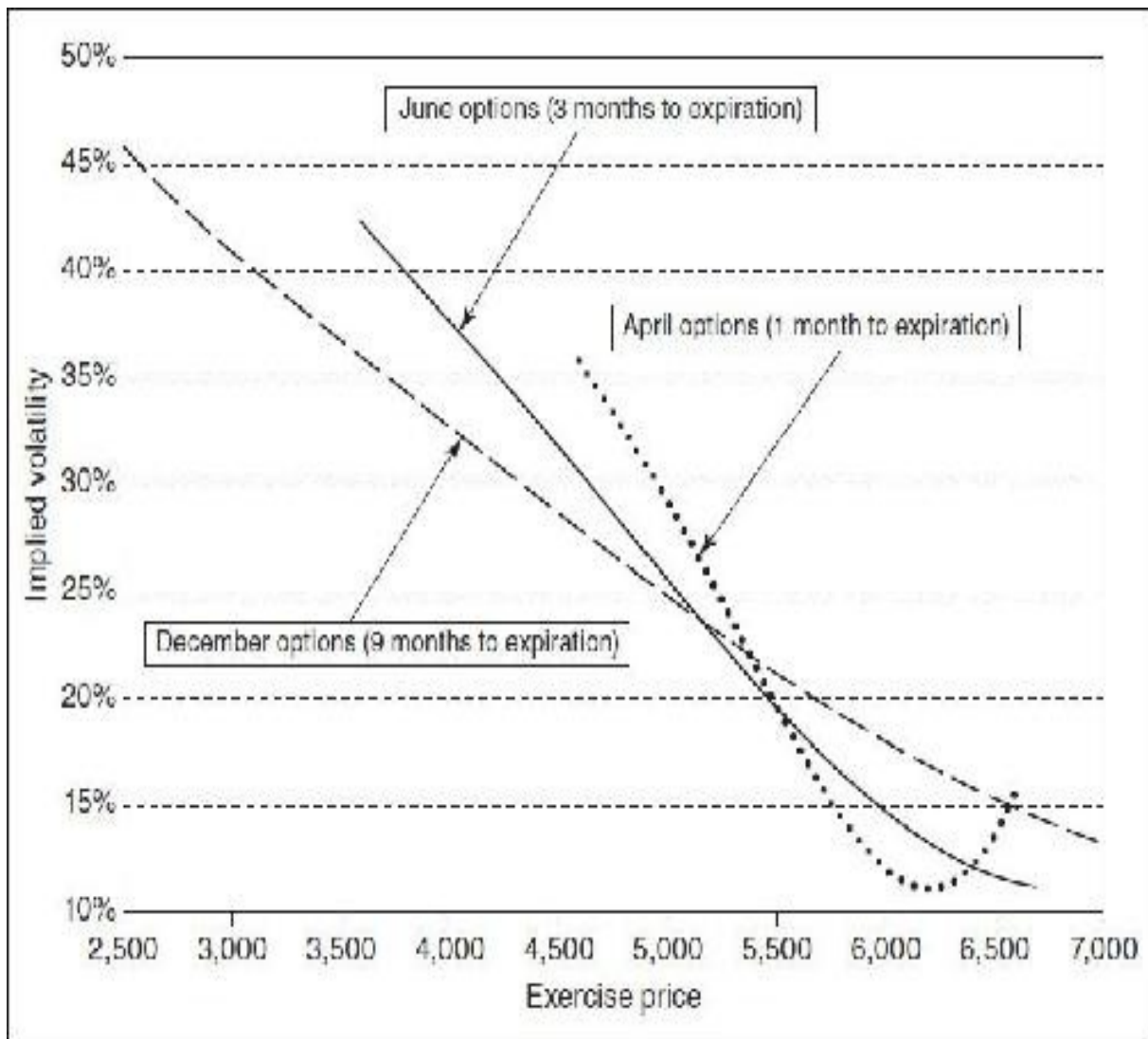
Shifting the entire skew might be a reasonable approach if a trader believes that the shape of the skew will remain unchanged regardless of changing market conditions. But is this likely? The implied volatilities at different exercise prices are likely to depend on how the marketplace views the likelihood of either larger or smaller moves in the price of the underlying contract. But all moves are relative with respect to both the underlying price and the time to expiration. In relative terms, a price change of 10.00 is greater with an underlying price of 100 (a 10 percent move) than with an underlying price of 200 (a 5 percent move). In the same way, a 10 percent move is greater over a one-week period than the same 10 percent move over a one-month period.

A first step in adjusting for the relative magnitude of price changes is to express each exercise price along the x axis in terms of its *moneyness*—how far in the money or out of the money the exercise price is as a percent of the

underlying price. The 90 exercise price with the underlying price at 100 will have a moneyness of 0.90. This is the same moneyness as the 180 exercise price with an underlying price of 200. We can make a further refinement by expressing each exercise price in logarithmic terms $\ln(X/S)$, where S is the underlying or spot price and X is the exercise price. This is consistent with the assumption that underlying prices are lognormally distributed.

How will the passage of time affect the shape of the volatility skew? Consider a 90 put with the underlying contract trading at 100. As time passes, in relative terms, the 90 put is moving further out of the money. In an investment skew, as the option moves further out of the money, its implied volatility will rise. In a sense, it is moving “up the skew.” This will cause the skew to appear more severe as time passes, with lower exercise prices carrying increasingly higher implied volatilities. Higher exercise prices may also be affected by the passage of time because an out-of-the-money call will also go further out of the money. Depending on the shape of the skew and where an exercise price falls along the skew, its implied volatility may rise, fall, or remain the same. If no adjustment is made, the effect of time passing on the FTSE 100 skew is shown in [Figure 24-5](#).

Figure 24-5 FTSE 100 option implied volatilities, March 16, 2012 (FTSE = 5965.58).



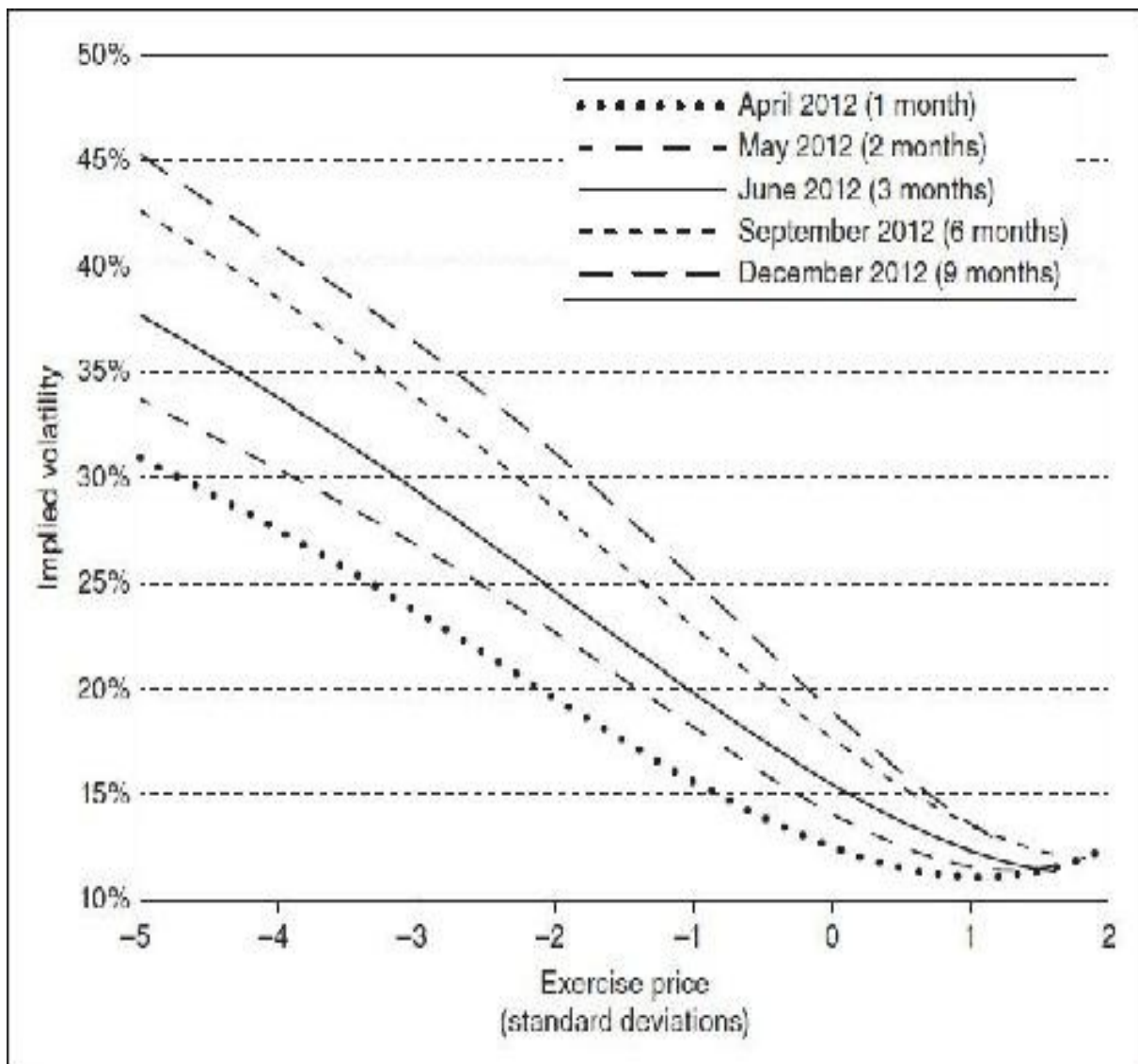
To compare volatility skews for different expirations, we need to determine theoretically how far in the money or out of the money an option is. Perhaps the easiest way to do this is to express each exercise price in terms of standard deviations away from at the money. Recalling the square-root relationship between time and volatility, and using our logarithmic scale, the number of standard deviations in the money or out of the money for each exercise price is given by

$$\frac{\ln(X/S)}{\sigma\sqrt{t}}$$

with an exactly at-the-money³ option having a standard deviation of 0. Skews for several FTSE 100 option expirations as of March 16, 2012, are shown in

[Figure 24-6](#). When expressed in this format, the skew is sometimes referred to as a *sticky-delta* skew because the delta is an approximation of how far in the money or out of the money an option is.

Figure 24-6 FTSE 100 implied volatilities, March 16, 2012.

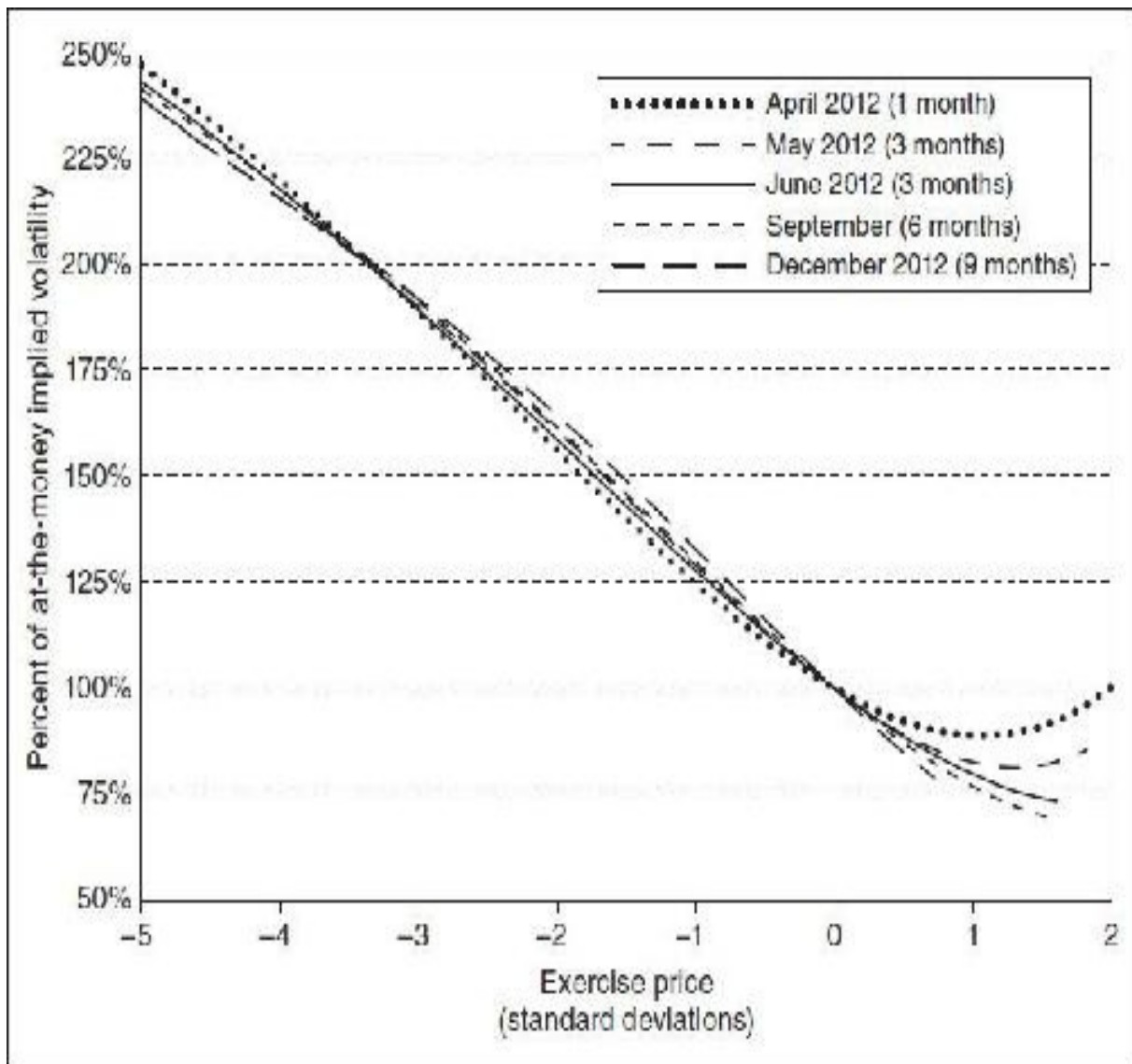


The skews in [Figure 24-6](#) appear to be similar, but they are clearly not identical. All adjustments thus far have been to the x axis, changing the calibration to more easily compare exercise prices. But we might also adjust the y axis, the volatility. When a trader refers to the overall implied volatility in a market, he is almost always referring to the implied volatility of at-the-money options. Whether the implied volatility at any exercise price is high or low will depend on whether it is high or low compared with the at-the-money implied

volatility. As a result, many traders recalibrate the y axis in terms of how the implied volatility at an exercise price compares with the at-the-money implied volatility. We can do this by expressing y values as the difference between the implied volatility of an at-the-money option and the implied volatility at each exercise price. If the at-the-money implied volatility is 20 (percent) and the implied volatility at an exercise price is 25 (percent), the y value is $20 - 25 = -5$. If the implied volatility at a different exercise price is 18 (percent), the y value is $20 - 18 = 2$.

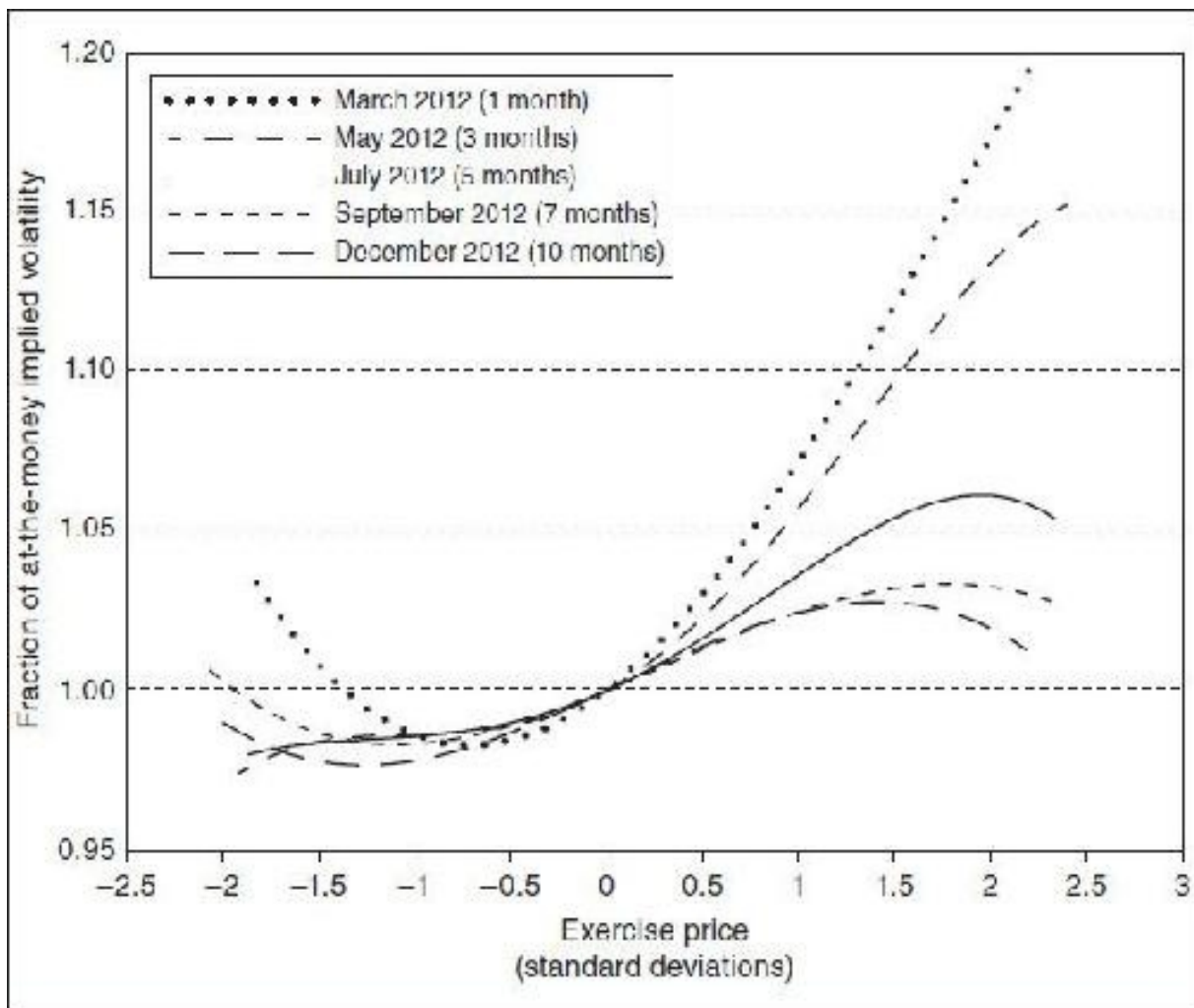
This method may be satisfactory if implied volatilities remain relatively constant, but suppose that the at-the-money implied volatility doubles from 20 to 40 percent. We might also expect the volatility at each exercise to double. An exercise price that previously had an implied volatility of 25 percent will now have an implied volatility of 50 percent, and an exercise price that previously had an implied volatility of 18 percent will now have an implied volatility of 36 percent. We can better calibrate the y axis by expressing the volatility at each exercise price as a percent of at-the-money implied volatility. With an at-the-money implied volatility of 20 percent, an implied volatility of 25 percent would be expressed as $25/20 = 125$ percent. An implied volatility of 18 percent would be expressed as $18/20 = 90$ percent. And an implied volatility equal to the at-the-money implied volatility would be expressed as $20/20 = 100$ percent. In [Figure 24-7](#), the y axis for the sample FTSE 100 skews has been recalibrated using this approach.

Figure 24-7 FTSE 100 implied volatilities, March 16, 2012.



[Figure 24-7](#) is typical of many stock indexes, exhibiting a very pronounced investment skew, with lower exercise prices significantly inflated compared with higher exercise prices. A different set of skews, for wheat options, is shown in [Figure 24-8](#). In this example, the skews exhibit more curvature but with higher exercise prices somewhat more inflated, as is often the case with a demand or commodity skew. The skews also seem to exhibit less consistency across different expiration months than the FTSE 100. While skews in a financial product tend to be similar across expiration months, skews in a commodity market can often vary across different expirations, perhaps owing to seasonal volatility considerations or because of short-term supply and demand imbalances.

Figure 24-8 wheat implied volatilities, January 27, 2012.



The foregoing method of modeling a skew is used by many traders but is in no way meant to be definitive. Adjustments are often required to prevent the model from generating illogical volatilities or theoretical values. For example, as we reduce volatility, an out-of-the-money option goes further out of the money because it is a greater number of standard deviations away from the underlying price. But in an investment skew, as a put goes further out of the money, its volatility is rising—it is “climbing the skew.” If the skew is sufficiently steep, the increase in volatility may in fact cause the theoretical value of the put to rise. This is inherently illogical because we expect all option values to decline if we reduce volatility.

Skewness and kurtosis

The shape of the skew is not constant. As market conditions change, option prices will also change, often causing the shape of the skew to change. Two common changes have to do with the tilt and curvature of the skew. The tilt, which defines how much the implied volatilities of lower exercise prices differ from the implied volatilities of higher exercise prices, is often referred to as the *skewness*. This follows logically from the definition of skewness in [Chapter 23](#) (see [Figure 23-10](#)). If the probability distribution has a longer left tail (negative skewness), there is greater likelihood of large down moves, resulting in greater demand for lower exercise prices. If the distribution has a longer right tail (positive skewness), there is greater likelihood of large up moves and, consequently, greater demand for higher exercise prices. Examples of positive and negative skewness are shown in [Figure 24-9](#).

Figure 24-9 Skewness.

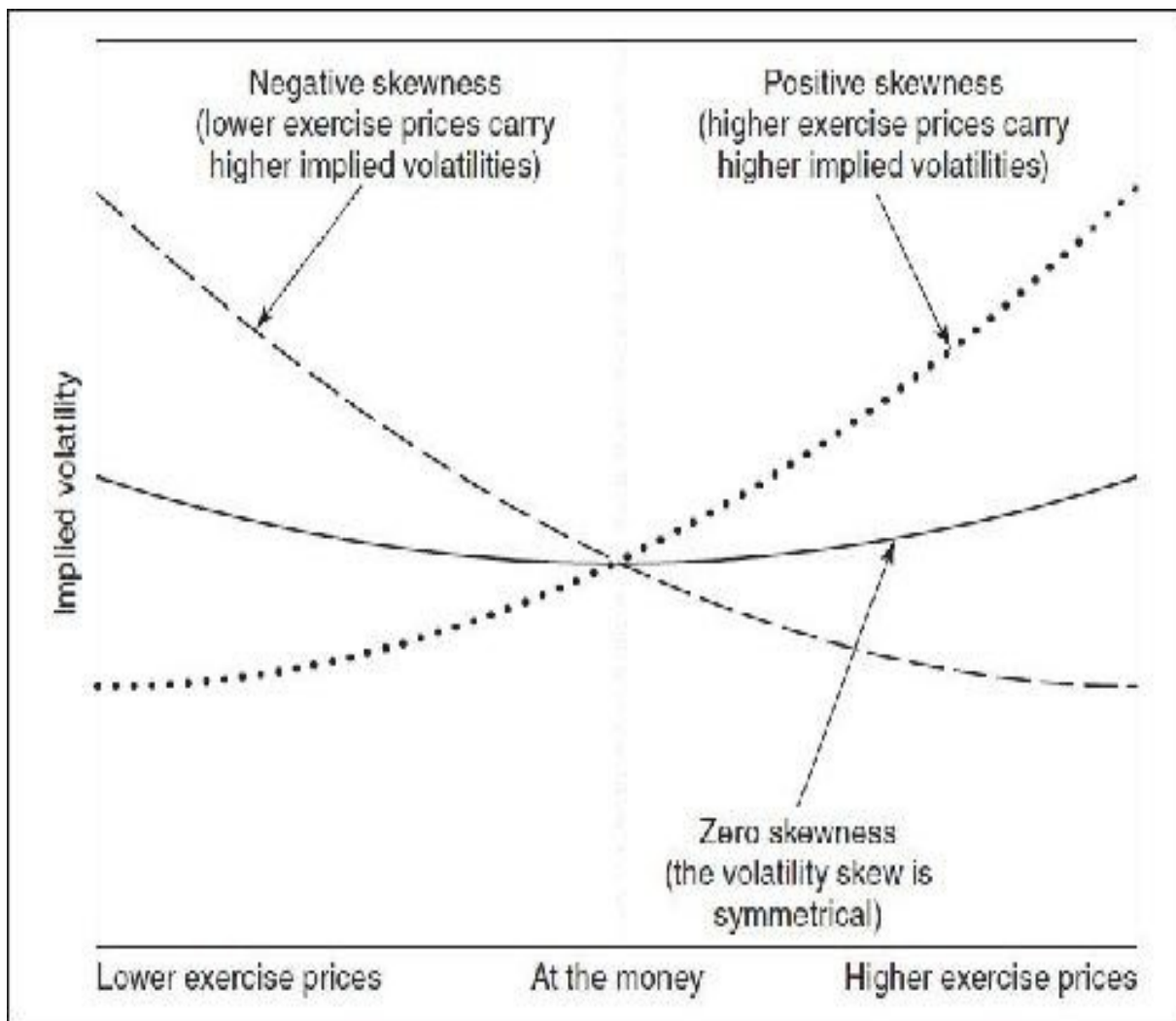
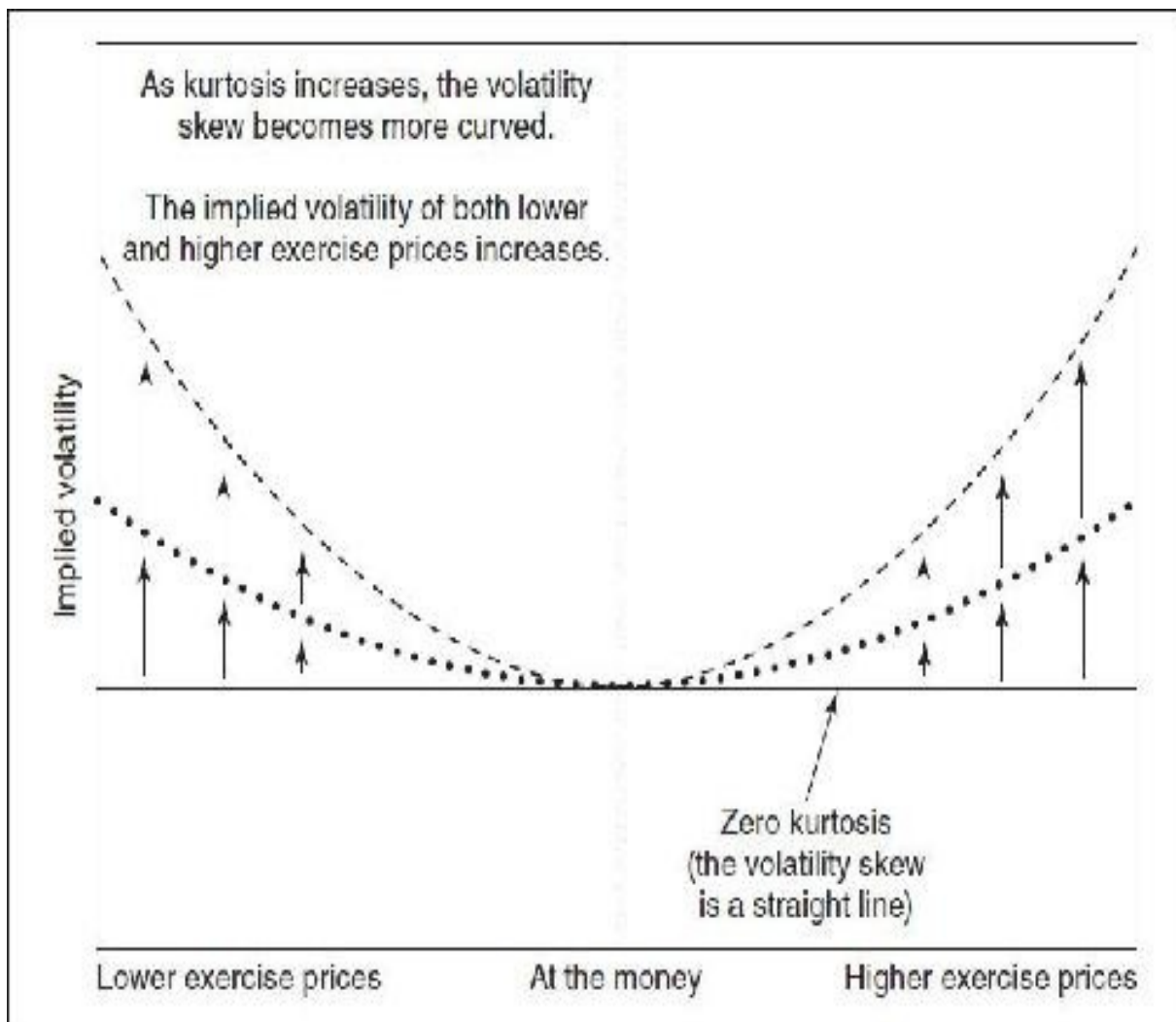
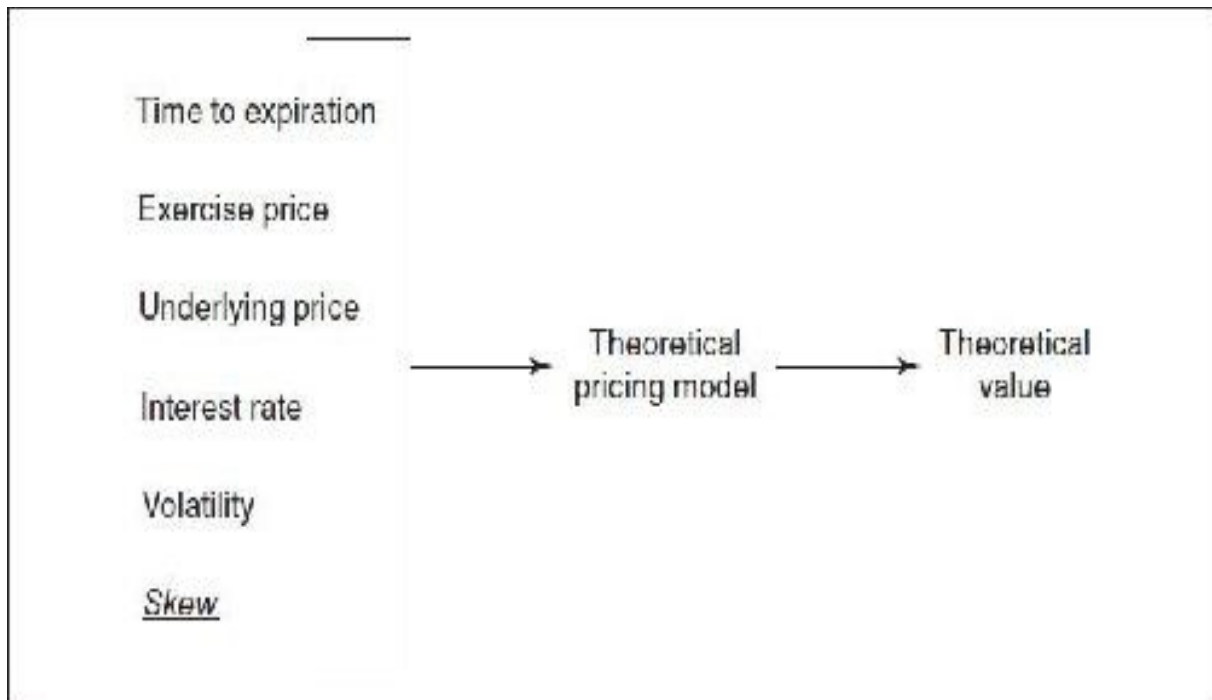


Figure 24-10 kurtosis.



The curvature, or *kurtosis*, defines how much the implied volatilities of both higher and lower exercise prices are inflated compared with the at-the-money implied volatility. This also follows logically from the definition in [Chapter 23](#) (see [Figure 23-11](#)). If the probability distribution has “fat tails,” there is a greater likelihood of large moves in either direction. Consequently, there will be a greater demand for out-of-the-money options (positive kurtosis). Examples of increasing positive kurtosis are shown in [Figure 24-10](#).⁴

Figure 24-11 The skew as a model input.



We might think of the skew as an input into a theoretical pricing model ([Figure 24-11](#)), but the skew is input into the model as a formula rather than as a single number. As with any input, it will be useful to determine how sensitive an option value or option position is to changes in the shape of the skew.

The sensitivities associated with a skew will depend on the skew model that is used. For example, let's assume a very simple second-degree-polynomial model where the volatility y at an exercise price x is given by

$$y = a + bx + cx^2$$

In this model, the value of a is the base volatility, usually the implied volatility of the at-the-money options. The values of b and c represent the skewness and kurtosis, respectively, of the volatility skew. We can raise or lower the value of a as implied volatility rises or falls. We can raise or lower the value of b to increase or decrease the skewness. And we can raise or lower the value of c to increase or decrease the kurtosis. b can be either positive or negative depending on whether higher or lower exercise prices are inflated. For exchange-traded markets, the value of c is almost always positive because the probability distributions of these markets always exhibit some fat-tail characteristics.

The sensitivity of an option's theoretical value to a change in skewness or kurtosis will depend on how the option's value changes as we raise or lower the

value of b and c . If raising the value of b by one unit will cause the option to fall by 0.15, then the option has a skewness sensitivity of -0.15 . If raising the value of c by one unit will cause the option to rise by 0.08, the option has a kurtosis sensitivity of 0.08. For active traders who carry very large option positions, the skewness and kurtosis sensitivities can represent significant risks and, as with all risks, must be monitored to ensure that they remain within acceptable bounds.

The units used to express skewness and kurtosis sensitivity will depend on how the skew model has been constructed. Most traders choose a unit that represents a common change in the skewness and kurtosis values. For example, if the value of b commonly ranges from 0.20 to 0.40, a logical unit for b might be 0.01. If the unit value is an unwieldy number, the value can be adjusted by including a multiplier. If the unit value for b is 0.001 but we wish to express the unit value as a whole number, we can use a multiplier of 0.001 to yield a unit value of 1. The model will then be expressed as

$$y = a + 0.001bx + cx^2$$

If we raise the skew value of b by 1, we are really raising it by 0.001. The same approach can also be used to express c in simple units.⁵

In most skew models, the at-the-money exercise price acts as a pivot point so that an option that is exactly at the money has a skewness and kurtosis sensitivity of 0. Options that are in the money or out of the money can have either a positive or a negative skewness sensitivity. If we increase the skewness input, the volatility of higher exercise prices will rise, while the volatility of lower exercise prices will fall. Consequently, higher exercise prices will have positive skewness sensitivity values, and lower exercise prices will have negative sensitivity. If we increase the kurtosis input, the volatility of options at both higher and lower exercise prices will rise. Consequently, any option that is not exactly at the money will have a positive kurtosis sensitivity.

Which options are the most sensitive to changes in skewness and kurtosis? There is no definitive answer because it depends on the volatility characteristics of the market as well as the skew model that is used. But in many skew models puts with deltas of -25 and calls with deltas of $+25$ tend to have the greatest skewness sensitivity. For this reason, a common measure of skewness is the difference between the implied volatility of the -25 delta put and the $+25$ delta call. There is no similar benchmark for kurtosis, but for many models, puts with

deltas of approximately -5 and calls with deltas of $+5$ tend to have the greatest sensitivity to a change in kurtosis.

Skewed Risk Measures

How we model the volatility skew also will affect the risk measures generated by a model—the delta, gamma, theta, and vega. Look again at [Figure 24-4a](#), where the floating skew is shifted either right or left as the underlying price rises or falls. As the skew is shifted, the volatility at some exercise prices will rise, while the volatility at other exercise prices will fall. This change in volatility can cause an option's value and its risk sensitivities to change either more or less than expected if there were no skew.

For example, consider an out-of-the-money put with a delta of -20 . Ignoring the gamma, if the underlying price rises 1.00 , we expect the option value to decline by 0.20 . But in an investment skew, such as in [Figure 24-4a](#), as the underlying price rises, the volatility of an out-of-the-money put will rise as it moves further out of the money. If the option has a vega of 0.10 and the shift in the skew causes the implied volatility of the option to rise 0.5 percent, the higher volatility will cause the option's value to rise by $0.5 \times 0.10 = 0.05$. Consequently, the option will only decline by 0.15 , a decline of 0.20 due to a change in the underlying price combined with an increase of 0.05 due to the increase in implied volatility. The option has a *skewed* or *adjusted delta* of -15 .

The inclusion of a volatility skew in a pricing model will affect the calculation of all option risk measures and can greatly complicate a trader's ability to manage risk. For many traders, it may be best to keep things simple, perhaps using a skew model to generate theoretical values while using a traditional model to calculate the delta, gamma, theta, and vega. For an active trader who carries large option positions, calculating accurate skewed sensitivities becomes much more important because the total value of the position can change very quickly as market conditions change. Financial engineers at professional option trading firms are often responsible for developing methods to accurately calculate theoretical values and risk sensitivities using a volatility skew model. But even the most sophisticated model is unlikely to generate values that exactly model option prices under all market conditions. A model can help, but it will always have limitations.

By combining the volatility term structure across expiration dates with the

volatility skew across exercise prices, we can form a *volatility surface*. While sometimes difficult to visualize, a volatility surface may enable a trader to more easily see the basic volatility characteristics of an option market. The more exercise prices and expiration dates that are available, the more accurate will be the volatility surface. Sample volatility surfaces for the FTSE 100 options and wheat options are shown in [Figures 24-12](#) and [24-13](#). At the time, the FTSE 100 Index was trading at 5,966, and the front-month wheat futures contract was trading at 647.

Figure 24-12 FTSE 100 volatility surface, March 16, 2012.

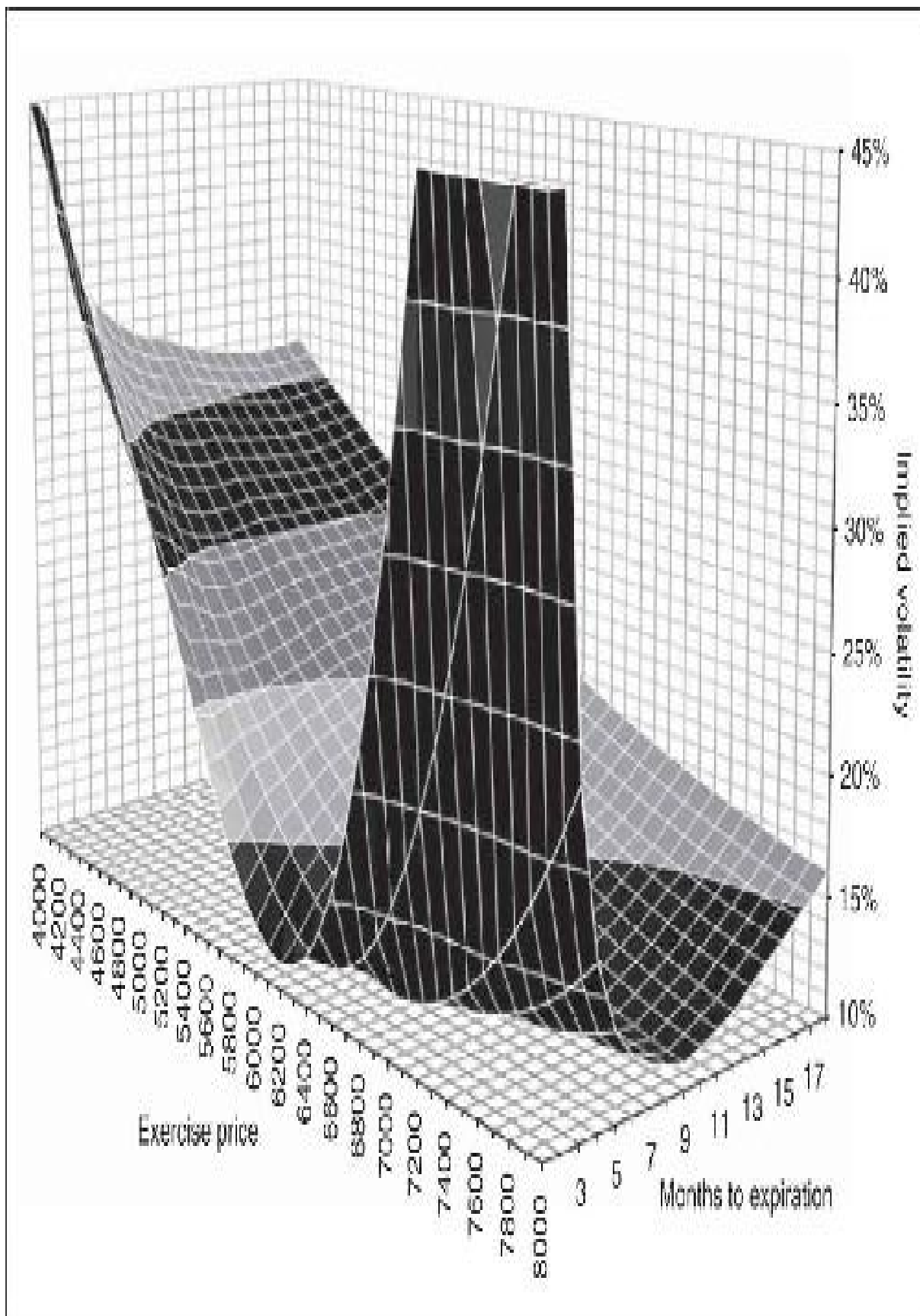
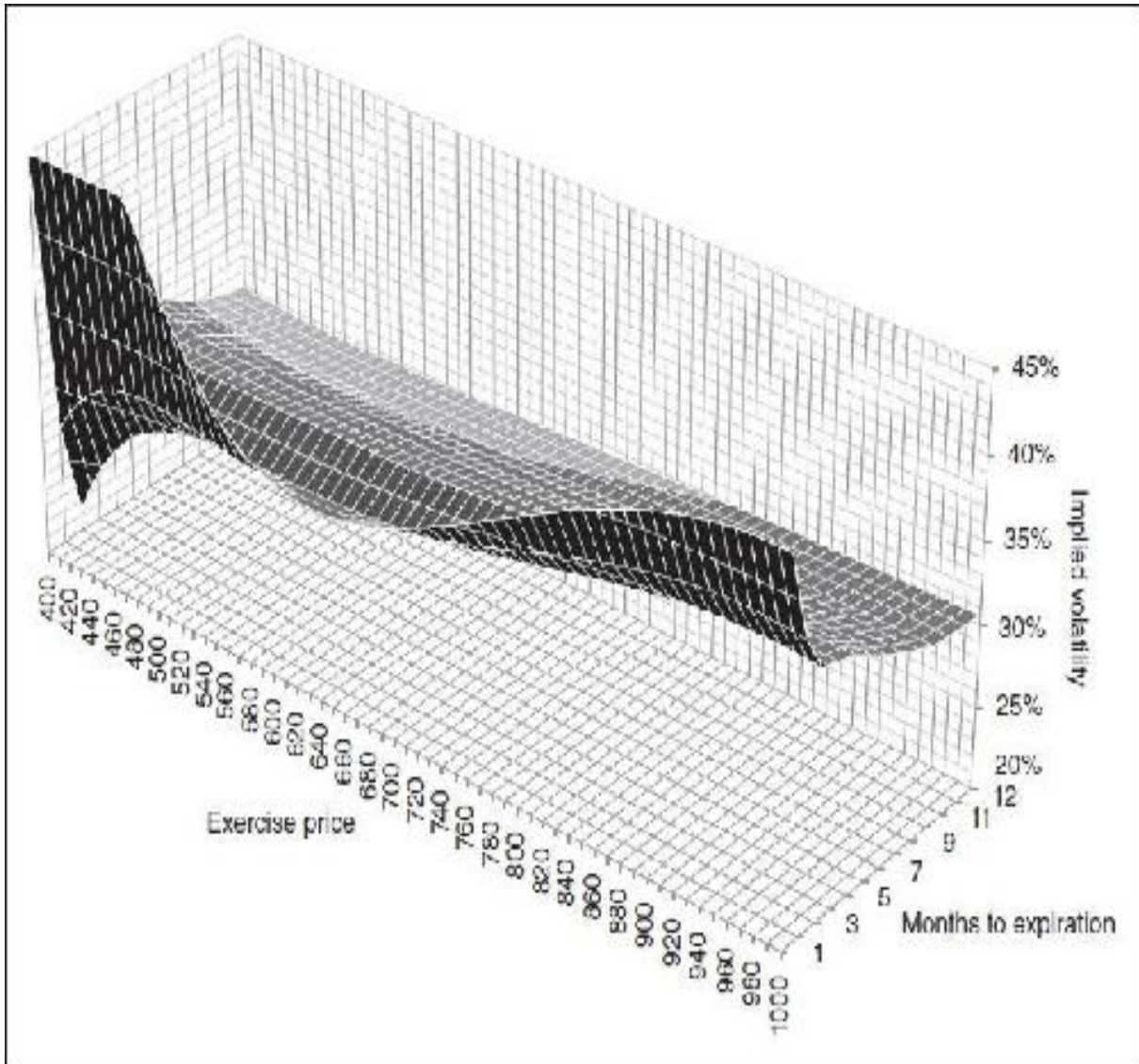


Figure 24-13 wheat volatility surface, January 27, 2012.



Shifting the Volatility

Traders have long noted that in many option markets, implied volatility tends to change as the price of the underlying contract changes. Some markets exhibit a direct relationship between movement in the underlying price and changes in implied volatility: when the underlying price rises, implied volatility tends to rise; when the underlying price falls, implied volatility tends to fall. This is typical of markets with a demand skew, such as agricultural and energy products. Other markets may exhibit an inverse relationship: when the underlying price

rises, implied volatility tends to fall; when the underlying price falls, implied volatility tends to rise. This is typical of markets with an investment skew, such as stock and stock index markets.

For purposes of both option evaluation and risk management, many traders will attempt to incorporate this characteristic into an option pricing model. One possible theoretical solution is the CEV model referred to in [Chapter 23](#). But this model can be mathematically complex and requires additional inputs, all of which make it difficult to use. Alternatively, many traders simply use a “home grown” model that shifts the volatility up or down in a way that is consistent with the observed volatility characteristics of a market. However, no model will generate accurate values under all conditions because implied volatility often changes in ways that seem to defy even the best model.

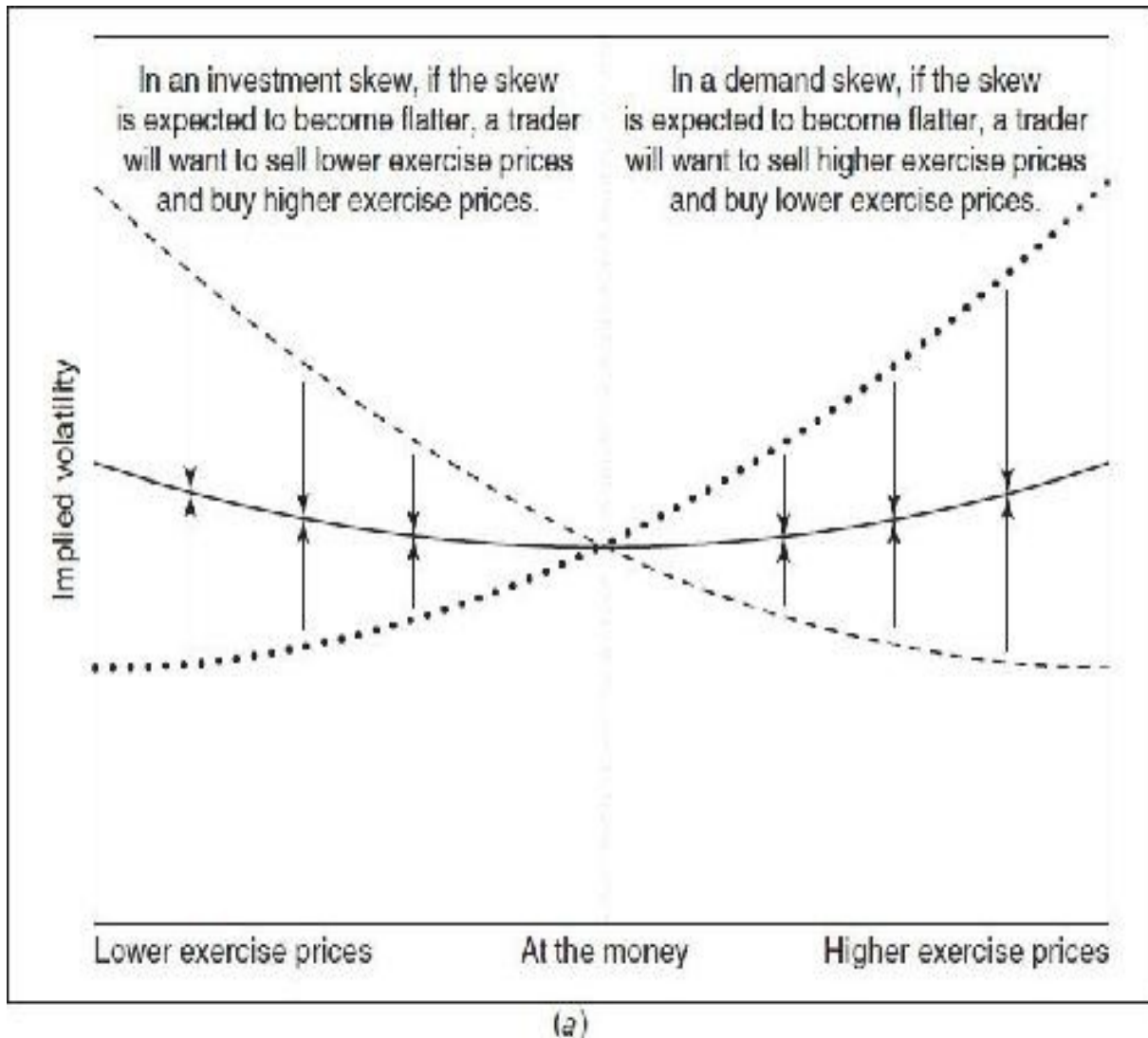
A shift in volatility can also affect the risks associated with a position. Consider a trader who buys an at-the-money straddle. Ignoring interest considerations and slight adjustments for a lognormal distribution, the trader’s position is approximately delta neutral: the call has a delta of 50, and the put has a delta of -50 . But delta neutral means that the trader has no particular preference for market movement in one direction or the other. Is this really true? If this position is taken in a stock index market, the trader actually has a preference for downward movement because he prefers higher volatility, something that is more likely to occur in a falling market. Even though the position may be delta neutral in a theoretical world, in the real world, it is delta negative. On the other hand, if this position is taken in a commodity market, where the market is likely to become more volatile when prices rise, the trader really has a positive delta position. Of course, it may be difficult to determine the real-world delta of either position. That will depend on how fast volatility rises or falls as the underlying price changes. But in neither case is the position truly delta neutral.

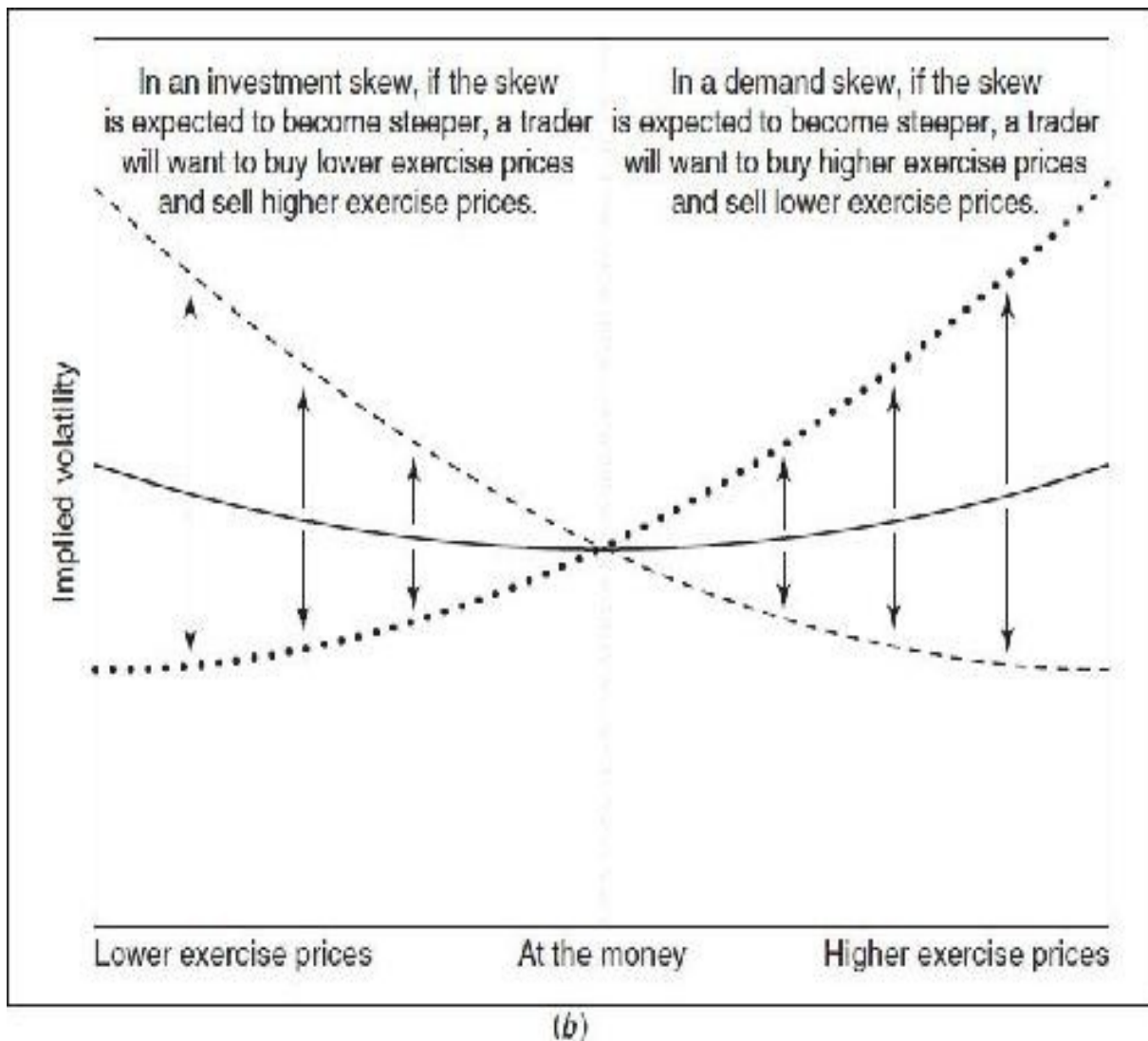
Skewness and kurtosis Strategies

Just as a trader may have an opinion about the direction of movement in a market (delta strategies) or about implied and realized volatility (vega and gamma strategies), a trader may also have an opinion about the shape of the volatility skew. We can see in [Figure 24-14](#) that, depending on the type of skew and whether a trader expects the skew to become steeper or flatter, the trader will

want to buy lower exercise prices and sell higher exercise prices, or vice versa. This is most commonly done using out-of-the-money options, very often 25 delta calls and -25 delta puts because these options tend to be most sensitive to changes in the slope of the skew.

Figure 24-14 (a) Declining skewness. (b) Increasing skewness.





If a skew trade is not hedged, the position will clearly have a positive delta (long calls and short puts) or negative delta (long puts and short calls). A trader who wants to focus solely on “buying skew” or “selling skew” must offset the delta position, most commonly with an opposing delta position in the underlying contract. When this is done, the entire strategy is usually referred to as a *risk reversal*. With the underlying contract trading at a price close to 100, the following are typical risk reversals (delta values are in parentheses):

+10 June 95 puts (−25)
 −10 June 105 calls (+25)
 +5 underlying contracts

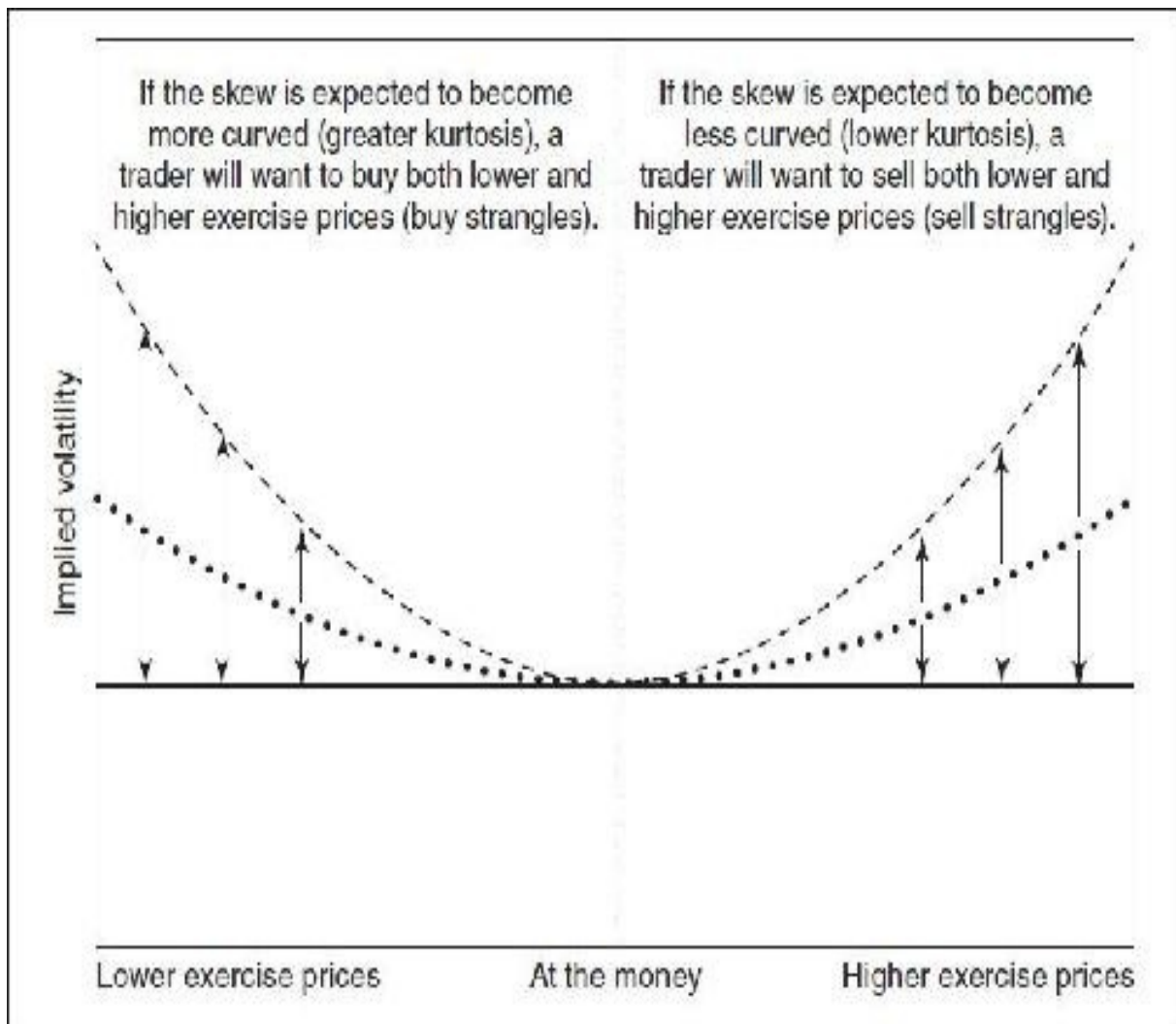
or

–30 December 90 puts (–15)
+30 December 110 calls (+15)
–9 underlying contracts

In these examples, the calls and puts have the same delta, but this is not a requirement. More commonly, calls and puts are chosen with the same vega values.⁶ This ensures that the position is vega neutral at inception and therefore primarily sensitive to changes in the slope of the skew rather than changes in overall implied volatility. Of course, as market conditions change, the delta, gamma, and vega of the position will almost certainly change. When this occurs, a trader will have to decide whether to maintain the position and, if so, how best to manage the delta, gamma, and vega risk. The risk characteristics of a typical risk reversal were discussed in [Chapter 21](#).

Just as a trader may have an opinion about skewness, the slope of a volatility skew, a trader may also have an opinion about kurtosis, the curvature of a volatility skew. If the kurtosis is expected to increase, the prices of options at both lower and higher exercise prices will increase. A trader will therefore want to buy strangles by purchasing both out-of-the-money calls and out-of-the-money puts. If the kurtosis is expected to decrease, option prices at both lower and higher exercise prices will decline. A trader will then want to sell strangles by selling both out-of-the-money calls and out-of-the-money puts. This is shown in [Figure 24-15](#).

Figure 24-15 Rising and falling kurtosis.



If a trader “buys” kurtosis by purchasing strangles or “sells” kurtosis by selling strangles, the position will also be sensitive to overall changes in volatility because the position will have a very pronounced positive or negative vega. Even if the trader is correct in his assessment of kurtosis, the position can be negatively affected by overall changes in implied volatility. If the trader wishes to focus solely on kurtosis, he will need to neutralize his vega position without changing the kurtosis of the position. Because at-the-money options are neutral with respect to kurtosis, a trader can achieve this by taking an offsetting vega position in at-the-money straddles. Assuming that the selected strangles and straddles are delta neutral, the entire position will also be delta neutral. With the underlying contract trading at a price close to 100, the following are typical kurtosis positions (vega values are in parentheses):

Long strangles:	+35 June 90 puts (0.11)
	+35 June 110 calls (0.12)
Short straddles:	−20 June 100 calls (0.20)
	−20 June 100 puts (0.20)

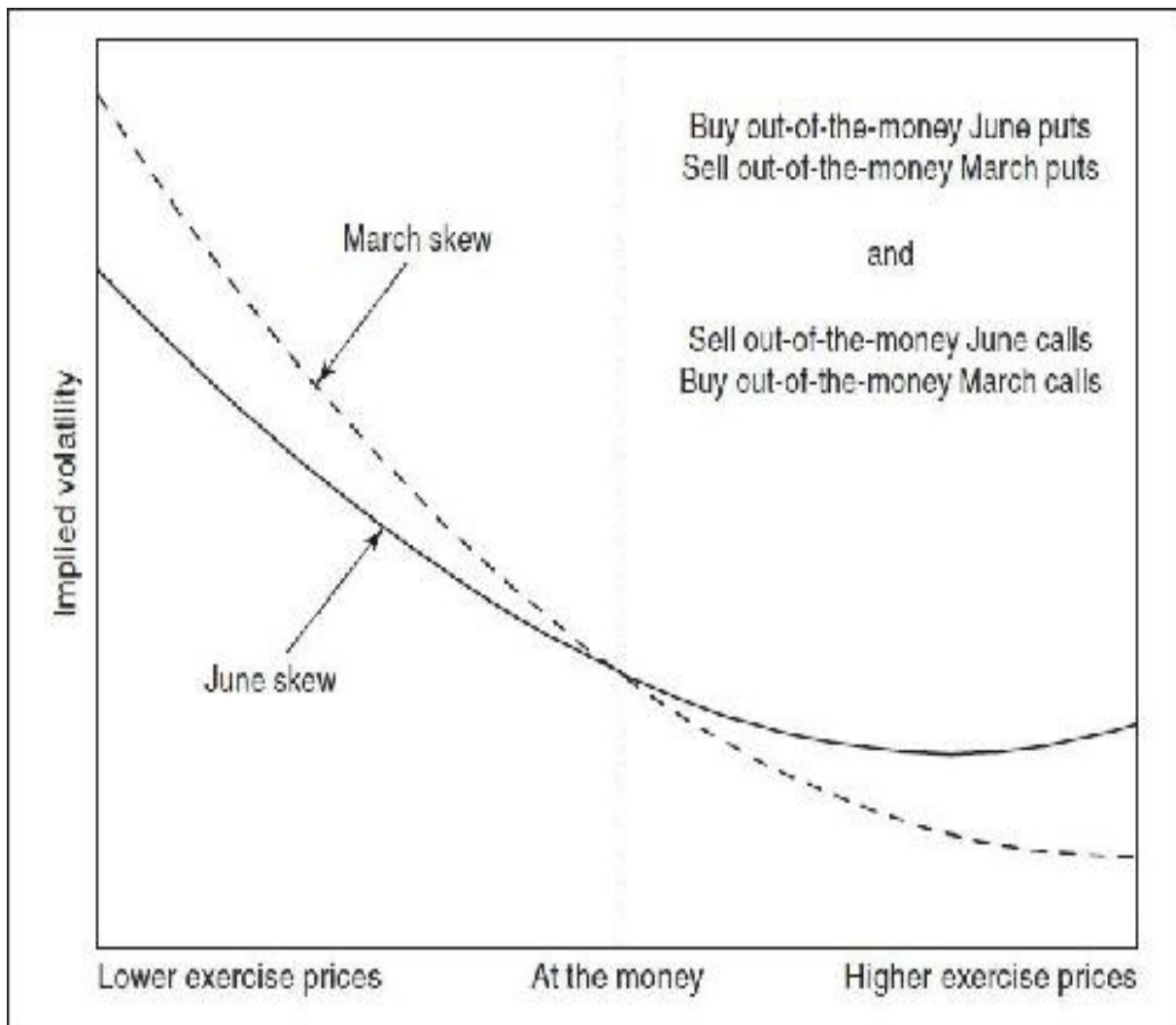
or

Short strangles:	−26 December 80 puts (0.08)
	26 December 120 calls (0.09)
Long straddles:	+10 December 100 calls (0.22)
	+10 December 100 puts (0.22)

If a kurtosis position is made up of a strangle that has exactly half the vega of the straddle, a vega-neutral position will consist of two strangles for each straddle. When done in this 2×1 ratio, the position is sometimes referred to as a *dragonfly*.

An opinion about skew and kurtosis can also be incorporated into other strategies. Consider the skews in [Figure 24-16](#) on the same underlying product but for different expiration months. If a trader has no opinion on whether either skew is mispriced individually but believes that the skews are mispriced with respect to each other, a logical strategy might be to take a skew position in one expiration month and an opposing skew position in the other month. For example, a trader might buy out-of-the-money puts in June and sell out-of-the-money puts in March. At the same time, the trader might sell out-of-the-money calls in June and buy out-of-the-money calls in March. The trader has, in effect, bought put calendar spreads and sold call calendar spreads. If the skew is the only consideration (the trader has no opinion on whether implied volatility is high or low), the trader will try to take a position that is vega neutral by choosing calendar spreads that have approximately the same vega. Any residual deltas can be hedged away with the underlying contract.

Figure 24-16 Buying and selling skew in different expiration months.

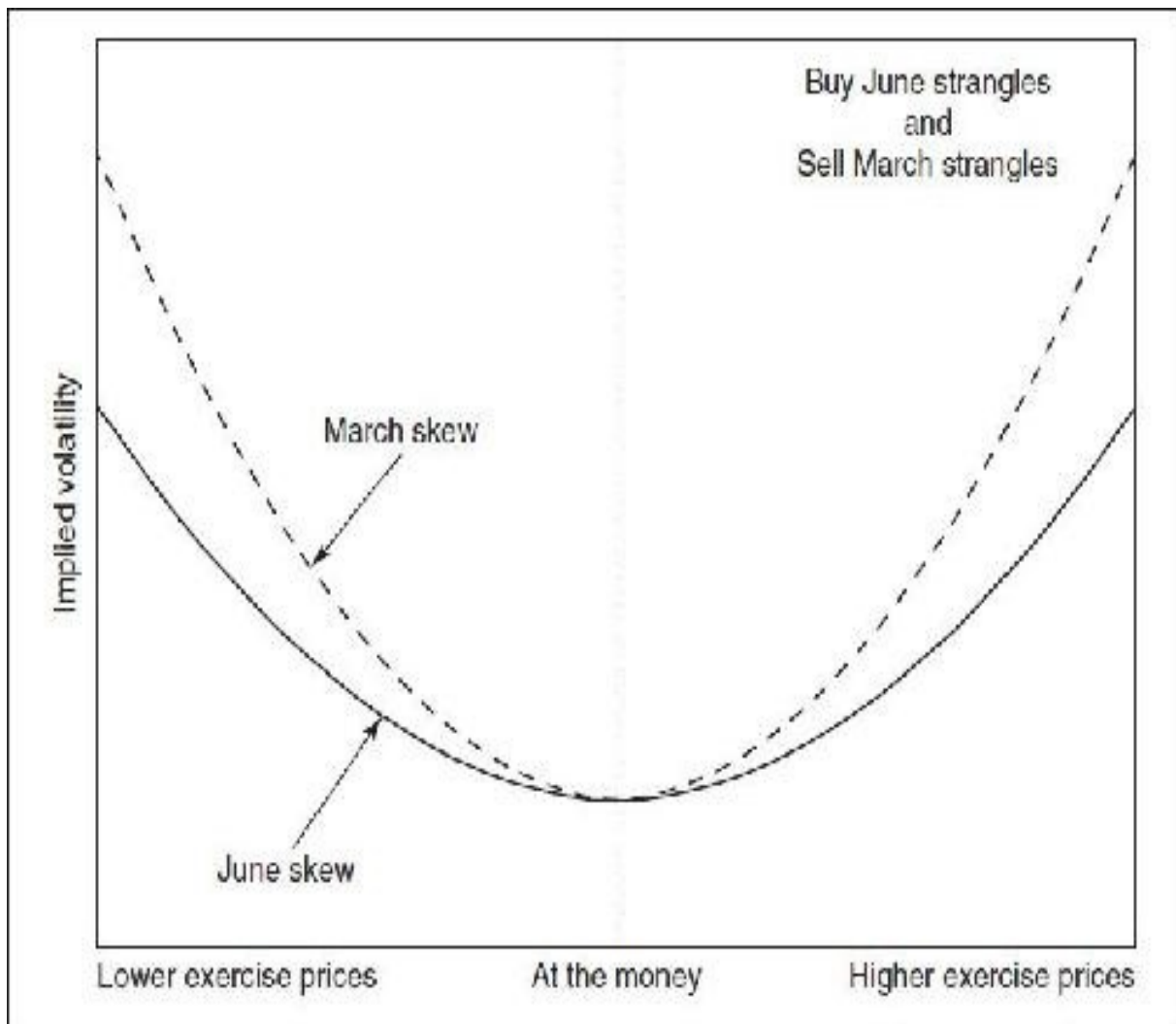


If, in addition to an opinion about the skew, a trader also has an opinion about the relative implied volatility in different expiration months, he can take this into consideration when choosing a strategy. If a trader believes that implied volatility in June is low compared with implied volatility in March, the trader will consider buying calendar spreads—buy June options and sell March options. If, at the same time, the trader also believes that the skews are mispriced with respect to each other, as in [Figure 24-16](#), he will choose calendar spreads that take both relationships into consideration. Now he will want to buy put calendar spreads—buy out-of-the-money June puts and sell out-of-the-money March puts. By doing so, the trader takes advantage of both implied volatility and skew. Note that the trader will avoid call calendar spreads because the volatility and skew will tend to offset each other. The June calls are too expensive with respect to skew, but the March calls are too expensive with

respect to implied volatility. If the trader believes that June implied volatility is high compared with March, now he will choose to sell call calendar spreads because the June calls are too expensive with respect to both implied volatility and skew.

The same approach can be used when kurtosis in two different expiration months seems to be mispriced, as shown in [Figure 24-17](#). Now a trader might consider buying June strangles and selling March strangles. If this is a simple kurtosis strategy within a single expiration month, it will be necessary to offset the vega by purchasing at-the-money straddles. But a trader can avoid this complication by choosing strangles in the two different expiration months that have approximately the same vega values. This ensures that the entire strategy is sensitive only to changes in kurtosis. If, at the same time, June implied volatility seems low compared with March, the strategy has an added advantage. June options are cheap compared with March options with respect to both volatility and kurtosis.

Figure 24-17 Buying and selling kurtosis in different expiration months.



Implied Distributions

In a perfect Black-Scholes world, the prices of the underlying contract are assumed to be lognormally distributed at expiration, and every option with the same expiration date ought to have the same implied volatility. The fact that options across different exercise prices have different implied volatilities must mean that the marketplace believes that the distribution of underlying prices at expiration is not lognormal. Exactly what probability distribution is the marketplace implying to the underlying contract at expiration? We can estimate this implied distribution by looking at the prices of butterflies in the marketplace.

At expiration, a butterfly has a minimum value of 0 if the underlying price

is at or outside the wings and a maximum value of the amount between exercise prices if the underlying price is exactly at the body, or midpoint, of the butterfly. At expiration, the 95/100/105 butterfly (i.e., buy a 95 call, sell two 100 calls, buy a 105 call) will have a minimum value of 0 if the underlying price is at or below 95 or at or above 105, a maximum value of 5.00 if the underlying price is exactly 100, or some amount between 0 and 5.00 if the underlying price is between 95 and 100 or between 100 and 105.

Suppose that exercise prices at five-point intervals are available extending from 0 to infinity:

..., 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130,...

What will be the value of the position at expiration if we buy every five-point butterfly?

...	+1 70 call	+1 75 call	...	+1 115 call	+1 120 call	...
...	-2 75 calls	-2 80 calls	...	-2 120 calls	-2 125 calls	...
...	+1 80 call	+1 85 call	...	+1 125 call	+1 130 call	...

Regardless of the underlying price at expiration, the entire position will always have a value of exactly 5.00. As a result, if we add up the prices of all the butterflies, the total value must be 5.00.⁷

Suppose that we make the assumption that the only prices that are possible at expiration are prices that are equal to an exercise price

..., 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130,...

The probability of each underlying price occurring must be equal to the price of that butterfly, where the inside exercise price is equal to the underlying price divided by 5.00. If the price of the 75/80/85 butterfly is 0.15, the probability of an underlying price of 80 at expiration must be

$$0.15/5.00 = 0.03 \text{ (3\%)}$$

If the price of the 90/95/100 butterfly is 0.50, the probability of an underlying price of 95 at expiration must be

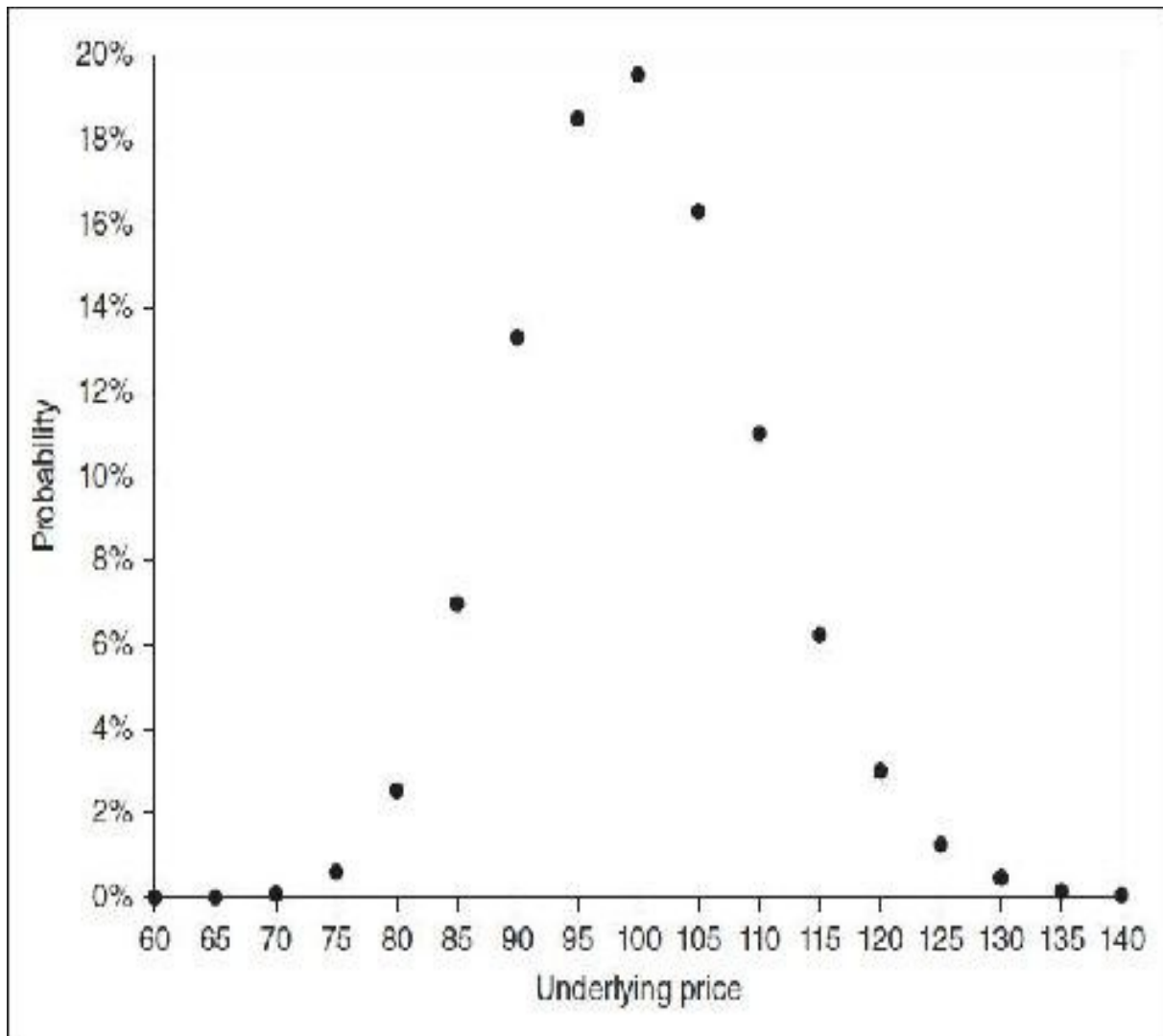
$$0.50/5.00 = 0.10 \text{ (10\%)}$$

[Figure 24-18](#) shows a series of call values together with the resulting butterfly values for our series of exercise prices.⁸ The probability associated with each underlying price is determined by dividing the butterfly value by the total value of all butterflies, which we know must be 5.00. (The reader may wish to confirm that all the butterfly values do indeed sum to 5.00 and that the probabilities sum to 1.00, or 100 percent.) The underlying prices and their associated probabilities are shown in [Figure 24-19](#). Note that these values form a probability distribution that is skewed to the right. This should come as no surprise because the values were derived from the Black-Scholes model, which assumes a lognormal distribution of underlying prices.

Figure 24-18 Butterfly values and probabilities.

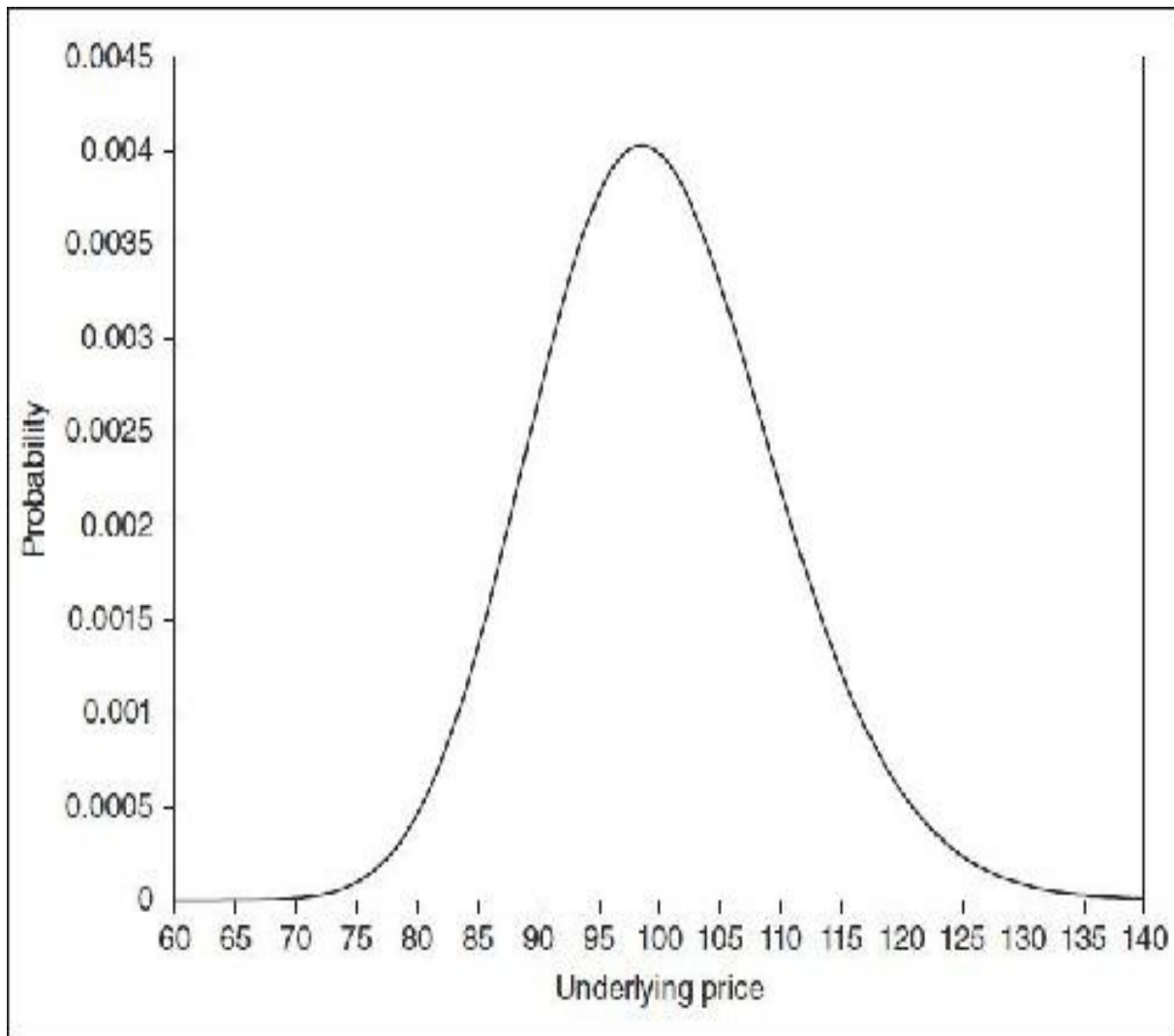
Exercise price:	75	80	85	90	95	100	105	110	115	120	125	130	135
Call value:	25.00	20.04	15.20	10.71	6.88	3.98	2.06	0.95	0.39	0.15	0.05	0.01	0
Butterfly value:	0.04	0.12	0.35	0.66	0.93	0.98	0.81	0.55	0.32	0.14	0.06	0.03	0.01
Price probability:	0.008	0.024	0.070	0.132	0.186	0.196	0.162	0.110	0.064	0.028	0.012	0.006	0.002

Figure 24-19 A discrete probability distribution implied from the prices of butterflies.



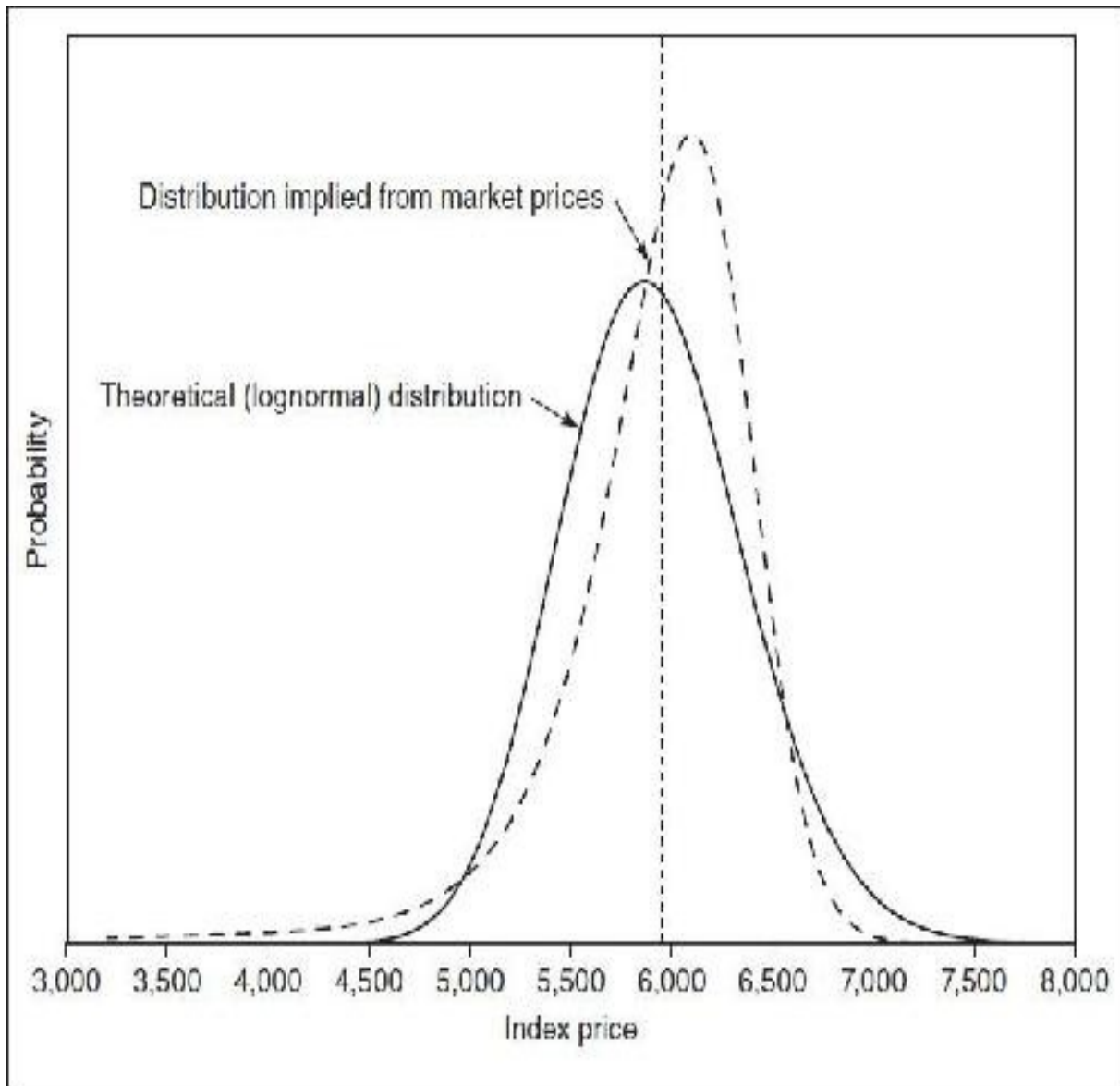
Of course, the distribution in [Figure 24-19](#) is only an approximation because it includes a limited number of underlying prices. A more exact distribution requires us to consider more and more exercise prices. We can do this by reducing the width of the butterflies. Instead of using increments of 5.00, we might use increments of 2.00, 1.00, or 0.50. Indeed, if we use increments that are infinitesimally small, the butterfly values will enable us to construct a continuous probability distribution. [Figure 24-20](#) shows the probability distribution with the increment between exercise prices reduced to 0.10. With such a small increment, the distribution appears almost continuous.

Figure 24-20 A continuous lognormal probability distribution implied from the prices of butterflies.



How does the distribution implied by option prices compare with a traditional lognormal distribution? The implied distribution will change as option prices change, so there cannot be one implied distribution under all market conditions. But we might get some sense of the distribution that the marketplace is implying by using the prices of butterflies generated by a volatility skew to derive a distribution and comparing it with a Black-Scholes distribution with a constant volatility. In [Figure 24-21](#), we have taken the volatility skew for the FTSE 100 options shown in [Figure 24-3](#) and created two distributions, one from the prices generated from the skew and one from prices generated from a constant volatility across every exercise price. What can we infer from [Figure 24-21](#)?

Figure 24-21 Three-month price distribution implied from FTSE 100 option prices, March 16, 2012 (FTSE 100 Index = 5,965.58).



Compared with a traditional lognormal distribution, the marketplace seems to be implying the following:

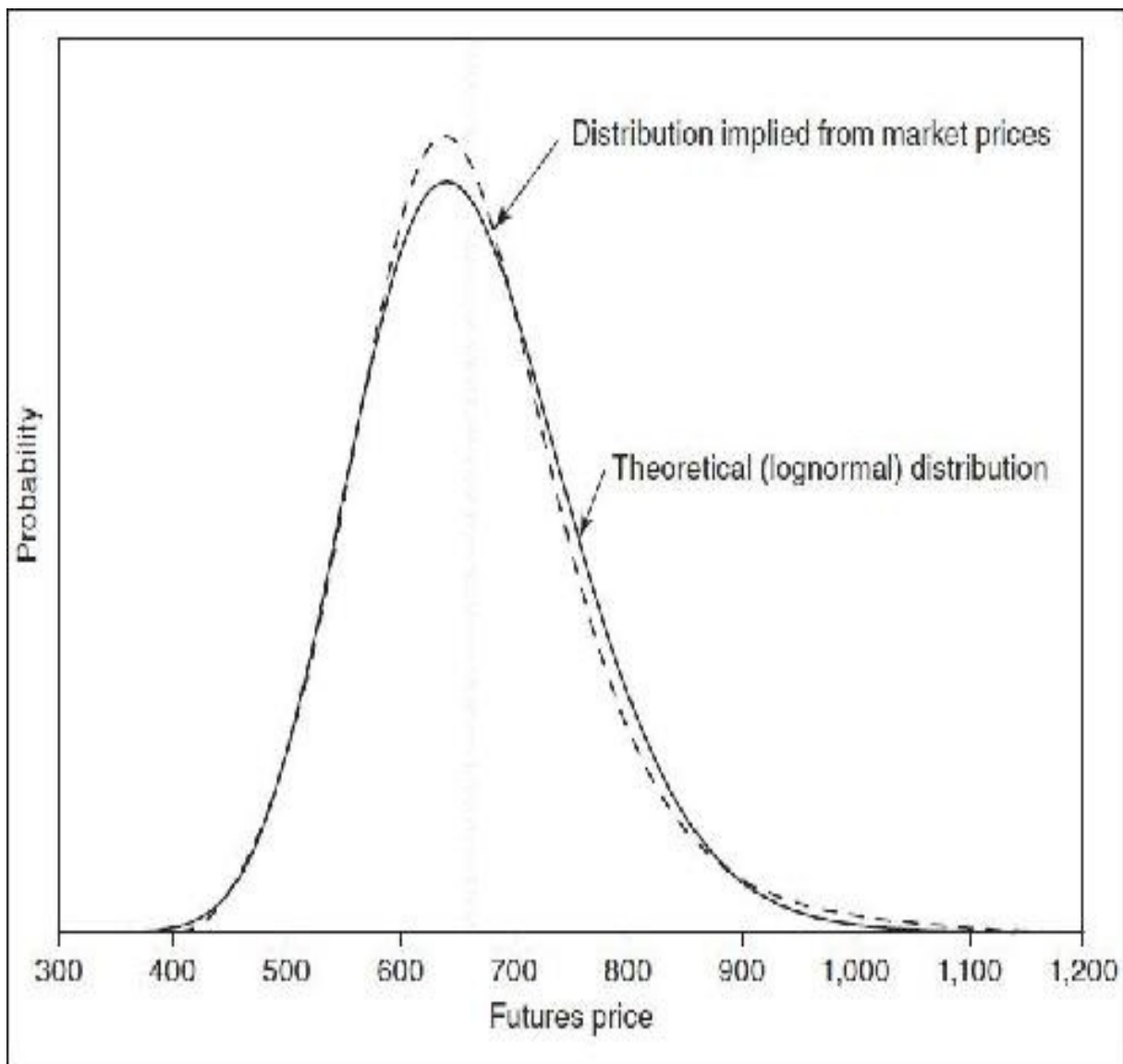
1. A greater probability of a small to intermediate upward move
2. A greater probability of a large downward move
3. A smaller probability of a small to intermediate downward move
4. A smaller probability of a large upward move

This implied distribution is typical of most stock index markets, and many of these points seem to be consistent with the S&P 500 histogram in [Figure 23-8a](#). There do seem to be more small moves in the real world than is predicted by a

theoretical distribution. There also seem to be more large downward moves and fewer intermediate downward moves. But the histogram also shows more big upward moves, which is not consistent with the implied distribution.

[Figure 24-22](#) shows the three-month distribution implied from the prices of options on wheat futures on January 27, 2012. This implied distribution seems to conform more closely to a theoretical lognormal distribution than does the distribution in the FTSE 100 example. However, the marketplace is still implying more small moves, slightly fewer intermediate upward moves, and slightly more large upward moves than a true lognormal distribution.

Figure 24-22 Three-month price distribution implied from wheat option prices, January 27, 2012 (with three-month wheat futures at 661.75).



Of course, [Figures 24-21](#) and [24-22](#) are snapshots of markets at one moment in time, and it would be unwise to draw any sweeping conclusions from these examples. Nonetheless, it can often be useful for a trader to compare his opinions about a probability distribution with that implied by prices in the marketplace. If there is a clear disagreement, it may point the way to a potentially profitable strategy.

¹ The Financial Times Stock Exchange 100 Index (the FTSE 100) is the most widely followed index of U.K. stock prices.

² There are, of course, investors and traders who take short stock positions, but they are relatively small in number compared with those who are long stock.

³ A more theoretically correct approach involves using the forward price F rather than the spot price, S

$$\frac{\ln(X/S)}{\sigma\sqrt{t}}$$

In this case, the at-the-forward option has a standard deviation of 0.

⁴ Because all exchange-traded markets seem to exhibit positive kurtosis, we ignore negative kurtosis skews.

⁵ When traders use the terms *skewness* and *kurtosis* (or *skew* and *kurt* for short), it is not always clear whether they are referring to the inputs into the model (the values of b and c in our example) or the sensitivity of the option's value to a change in these inputs. Typically, a trader will refer to the sensitivities as the option's skewness or kurtosis. Or the trader will refer to his skewness and kurtosis position: the sensitivity of his entire position to a one-unit change in the skewness or kurtosis inputs.

⁶ A risk reversal that is vega neutral will tend to be gamma neutral, although this will not always be the case. A trader may have to decide whether it is more important for the risk reversal to be vega neutral or gamma neutral.

⁷ If we include interest rates, and the options are subject to stock-type settlement, the total will be the present value of 5.00.

⁸ The values in [Figure 24-18](#) correspond, approximately, to Black-Scholes values using an underlying price of 100, three months to expiration, a volatility of 20 percent, and an interest rate of 0.